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From Healthy to Zombie

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Abstract

This report documents the development of *From Healthy to Zombie*, a single-player virtual reality (VR) survival simulation game set within a surreal supermarket environment. The project explores how VR can foster emotional engagement and emergent player behavior through physical interaction and environmental storytelling, without relying on traditional narratives or explicit violence. Core features include a physics-based object interaction system, lightweight AI-driven threats, a modular cutscene framework, and a differentiated scoring mechanism encouraging strategic decision-making. Emphasizing minimalistic design and player agency, the final deliverable demonstrates the potential of VR to create emotionally resonant, open-ended experiences within familiar yet distorted spaces. Key challenges encountered include managing system stability within a physics-heavy environment and balancing realism with playability. The project offers insights into emergent gameplay design, VR interaction principles, and ethical considerations for immersive media.

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Code link

GitHub source code: <https://github.com/thereisnouseusername1/COMP3000-Final-Year-Project/tree/development>

Trello board: <https://trello.com/b/ostosJ4D/3000-final-year-project>

Development video log:

<https://youtube.com/playlist?list=PLHmzA1o2pe8BMsebeBKRGgm6YjdQcTlsl&si=1amIE3fGyDSUz23N>

Downloadable file link: <https://thereisnouseusername1.itch.io/from-healthy-to-zombie>

Chapter 1: Introduction

Virtual reality (VR) technology has emerged as one of the most transformative mediums in modern digital entertainment, enabling players to experience environments and interactions with an unprecedented level of immersion.

Unlike traditional gaming platforms, VR bypasses the abstraction of input devices, allowing users to physically engage with virtual spaces through body movement, spatial awareness, and tangible interactions.

This project explores the potential of VR to deliver not just conventional gameplay mechanics but also **emotional and emergent experiences** through environmental design and physical interaction.

The primary focus of this work was the development of a VR-based game set within a supermarket environment—an everyday, familiar setting deliberately chosen to contrast with the surreal and unsettling gameplay experience.

The objective was to create a **sandbox-like world** where players could:

- Interact naturally with everyday objects.
- Face unorthodox threats (in the form of slow, rolling humanoid figures).
- Derive emotional responses through physical engagement and emergent narrative rather than scripted events.

By combining **physics-driven interaction**, **minimalist visual storytelling**, and **dynamic player agency**, this project aimed to demonstrate how VR can evoke complex emotions without relying on traditional dialogue, linear missions, or conventional combat systems.

In doing so, the project investigates the question:

How can virtual reality environments be designed to allow physical freedom and emergent narrative, while maintaining emotional impact and tension, within a familiar yet distorted space?

Chapter 2: Background, Objectives & Deliverables

2.1 Background

As virtual reality continues to expand into mainstream entertainment and education, there has been a growing exploration of how VR can deliver not just traditional game experiences but also new forms of emotional engagement.

In particular, environmental storytelling—the use of space, objects, and atmosphere to convey meaning—has become an increasingly powerful tool within VR.

While many VR games emphasize clear goals (e.g., combat victories, mission completion), there exists significant potential for experiences that emerge from player interaction and subtle environmental cues rather than explicit narratives.

This project was designed to explore that potential within a supermarket setting—an environment universally familiar yet capable of being transformed into something unsettling and surreal through careful design choices.

Previous titles such as Boneworks and Blade & Sorcery have demonstrated the power of physics-driven interactions in VR, while games like The Stanley Parable and Job Simulator illustrate the narrative possibilities when players are left free to interpret and experiment with their surroundings.

However, few existing VR experiences focus on creating emotional tension without overt violence, horror tropes, or linear storytelling.

This identified a gap which this project sought to investigate.

2.2 Project Objectives

The primary objectives of the project were as follows:

- **To design and develop a VR experience that emphasizes natural physical interaction over abstract mechanics.**
- **To create an emotionally charged environment using environmental storytelling without relying on scripted dialogue or traditional horror tropes.**
- **To enable player-driven emergent gameplay**, allowing freedom of action within a supermarket environment filled with interactive objects and dynamic threats.
- **To implement lightweight but scalable AI behavior** capable of maintaining tension without complex animations or navigation meshes.
- **To manage project development following professional standards**, including iterative testing, version control, and ethical considerations for VR gameplay.

2.3 Deliverables

The project aimed to deliver the following tangible outputs:

Deliverable	Description
VR Game Prototype	A functional VR application playable on Oculus Quest 2, featuring supermarket environment, physics-based interaction, and dynamic enemies.
Physical Interaction System	Grab, throw, and melee combat mechanics allowing players to manipulate everyday objects naturally.
Enemy AI Behavior	Basic enemy AI that pursues the player using grounded rolling movement, reacts dynamically to physical impacts, and recovers after explosive forces.
Cutscene and Environmental Event System	Scripted but player-triggered environmental changes (e.g., lighting shifts, enemy spawn events) to support atmospheric storytelling.
Final Year Project Report	A comprehensive academic report documenting design rationale, technical implementation, evaluation, reflection, and ethical considerations.

Chapter 3: Literature Review

3.1 Introduction to Relevant Works

To ground this project within existing research and practice, a number of significant virtual reality games and design principles were reviewed.

The focus was to identify how leading projects approached **physical interaction**, **environmental storytelling**, and **emergent gameplay** within VR or related spaces.

Three major strands of influence were identified:

- Physics-driven player interaction
- Non-linear or emergent narrative design
- Atmospheric tension without reliance on explicit horror

3.2 Key Game Studies

Boneworks (Stress Level Zero, 2019)



"Physics-based object manipulation in Boneworks (Source: Stress Level Zero)"

Boneworks is considered a landmark VR title for its commitment to full-body physical interaction.

Every object in the game adheres to a consistent physical system, allowing players to interact with the environment in highly emergent ways—stacking objects, blocking enemies, or solving problems creatively.

This inspired the current project's decision to:

- Use a **universal physics-based interaction system** (no special-case interactions).
- Allow all objects (food, weapons, props) to be picked up, thrown, or used naturally.

Furthermore, *Boneworks* highlighted the importance of consistent physics logic in maintaining player immersion, even at the cost of occasional awkwardness.

Blade & Sorcery (WarpFrog, 2018)

Blade & Sorcery focused on satisfying physical combat, where weapon collision and character reactions were entirely physics-driven.

Although more combat-oriented, it demonstrated how **moment-to-moment impact feel** could define the entire player experience.

From this, the project adopted:

- A focus on **responsive melee impact** without relying on animation triggers.
 - Implementation of **explosive reactions** using Unity's physics engine, ensuring hoomans react dynamically to player attacks.
-

Job Simulator (Owlchemy Labs, 2016)



"Example of playful object interaction in Job Simulator (Source: Owlchemy Labs)"

While totally different, *Job Simulator* exemplified how mundane environments (offices, kitchens) could be transformed into engaging VR playgrounds through exaggerated physics and freeform interaction.

This influenced the choice of a **supermarket** setting—taking a familiar, non-threatening space and reinterpreting it into something surreal through mechanics rather than narrative exposition.

The Stanley Parable (Galactic Cafe, 2013)

Although not a VR title, *The Stanley Parable* was crucial in understanding **environmental storytelling and player choice** without explicit objectives.

Players were free to interpret the meaning of their actions within seemingly absurd systems.

This philosophical approach informed the supermarket project's lack of enforced goals:

- Players may choose to fight, flee, explore, or do nothing.
- The game never explicitly rewards or punishes behavior, leaving interpretation to the player.

3.3 Academic and Technical Literature

Academic guidelines from Unity Technologies and VR development best practices, particularly regarding:

- Player comfort (reducing VR sickness)
- Physics optimization (managing Rigidbody interactions)
- Ethical considerations in immersive design (player agency vs. scripted manipulation)

were referenced throughout development to maintain professional standards.

Technical articles on **VR motion safety**, **object grabbing physics**, and **haptic feedback design** were reviewed, although full haptic implementation remained beyond scope for this prototype.

3.4 Gaps Identified

Despite the strengths of the games reviewed, few VR experiences combined:

- A mundane real-world environment
- Freeform, physics-driven gameplay
- Emergent emotional responses without overt narrative

This project aimed to fill that gap by offering a world where **emotion, tension, and interpretation arise organically from physical interaction and environmental context**.

3.5 Technologies

3.5.1 Visual Style Implementation: Cartoon Aesthetic

In order to establish a distinctive tone that balanced tension with accessibility, the game deliberately adopted a **cartoon-like visual style** rather than pursuing hyper-realistic graphics.

This aesthetic was achieved through a combination of technical techniques:

- **Low-Polygon Modelling:**
All environmental props, enemy models, and player-interactive objects were created using simplified geometric forms.
This maintained clarity in VR space, minimized visual noise, and reduced cognitive load during fast interactions.
- **Flat Lighting and Soft Shadows:**
Lighting was designed to avoid harsh contrasts or photorealistic reflections.
Ambient lighting with subtle, soft-edged shadows was used to create a slightly surreal, "plastic" appearance that enhanced the feeling of inhabiting an uncanny supermarket world.
- **Material Choices:**
Most surfaces used basic non-reflective materials with simple color palettes.
Specular highlights and normal mapping were intentionally minimized to reinforce the flattened, animated look.
- **Shader Simplicity:**
No custom toon shaders were used, preserving optimization for Oculus Quest 2 hardware, but Unity's Standard Shader was adapted with desaturated color settings and smooth materials to emulate cartoon aesthetics.

This combination produced a world that felt tangible and navigable while maintaining a slight detachment from reality—an essential balance for the project's emotional goals.

3.5.2 Influence of Other Works

Several existing titles influenced the decision to adopt a cartoon-inspired aesthetic:

- **Job Simulator (Owlchemy Labs, 2016):**
Demonstrated how exaggerated, non-realistic environments could encourage playful interaction and reduce the intimidation factor often associated with VR manipulation tasks.
- **Boneworks (Stress Level Zero, 2019):**
Although more physically realistic, Boneworks showed that **coherent internal logic** in object behavior was more critical for immersion than pure graphical fidelity. This reinforced the decision to prioritize consistent object physics over detailed visual textures.
- **The Stanley Parable (Galactic Cafe, 2013):**
Inspired the use of minimalist environmental design to provoke unease and player interpretation without explicit narrative delivery.

Thus, while the visual presentation was simplified, it drew upon sophisticated lessons in player perception, tone-setting, and environmental semiotics.

3.5.3 Narrative and Emotional Impact

The cartoon aesthetic was not merely a technical or stylistic choice; it served a deliberate narrative and emotional function:

- **Mitigation of Psychological Distress:**
By avoiding hyper-realistic depictions of humanoid enemies and physical violence, players could engage with the game space without significant emotional trauma, aligning with ethical VR best practices.
- **Enhancement of Absurdist Tone:**
The incongruity between setting (supermarket), threat (rolling humanoids), and aesthetic (cartoon softness) heightened the sense of surreal absurdity—a central theme of the experience.
- **Player Agency Amplification:**
A less visually oppressive environment encouraged exploration, experimentation, and emergent play, allowing players to project their own narratives onto the world rather than feeling directed.

Overall, the adoption of a cartoon aesthetic was instrumental in achieving the project's core goals of **player-driven emotional response, surreal tension, and ethical interaction design**.

Chapter 4: Method of Approach

4.1 Development Methodology

The project adopted an **iterative, modular development approach**, informed by principles from Agile software development but adapted for solo work within the constraints of an academic timeline.

Rather than adhering to rigid planning, the workflow emphasized:

- Rapid prototyping
- Immediate physical testing
- Frequent revision cycles based on playability and user feedback

Each major feature (e.g., enemy AI, physics interaction, environmental triggers) was treated as an independent module, allowing for:

- Easier debugging and isolated testing
- Parallel development without complex dependencies
- Greater flexibility in responding to unexpected challenges or opportunities

Short, self-imposed development sprints (typically one to two weeks) were used to maintain momentum and ensure continuous progress.

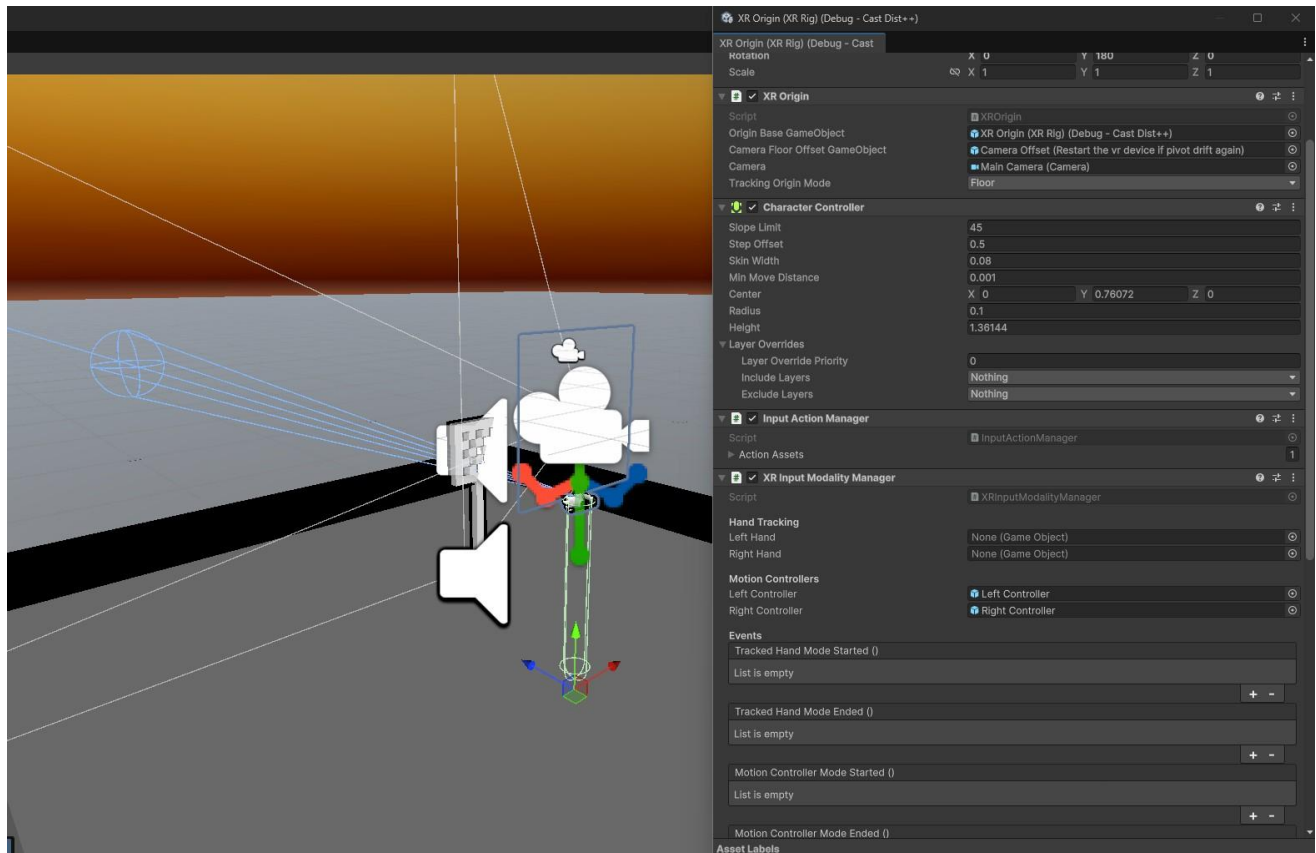
4.2 Core Design Principles

Several key principles underpinned all design decisions:

Principle	Description
Physicality First	All player interactions should derive from natural physical behavior (grabbing, throwing, hitting).
Minimalist Narrative	No explicit missions or dialogues; meaning should arise from environment and player choice.
Player Agency	Players should be free to engage or disengage with enemies and objects without forced objectives.
Consistency of Physics	Object behavior should follow universal rules to maintain immersion, even if it creates occasional awkwardness.
Performance Awareness	Given VR's resource demands, all assets, scripts, and physics systems were optimized for efficiency.

4.3 Technical Implementation Overview

4.3.1 XR Rig Setup



"Custom XR Rig prefab structure adapted for Oculus Quest 2"

The project used Unity's **XR Interaction Toolkit** with an XR Origin Rig, ensuring standardized head tracking, hand tracking, and interaction anchors.

Modifications included:

- Custom hand colliders to improve grabbing precision
- Wrist-mounted menu system designed to move along a cylindrical surface following hand rotation

Special care was taken to avoid common XR pitfalls, such as camera drift, through consistent anchoring and player height recalibration.

4.3.2 Player-Centric Design Principles

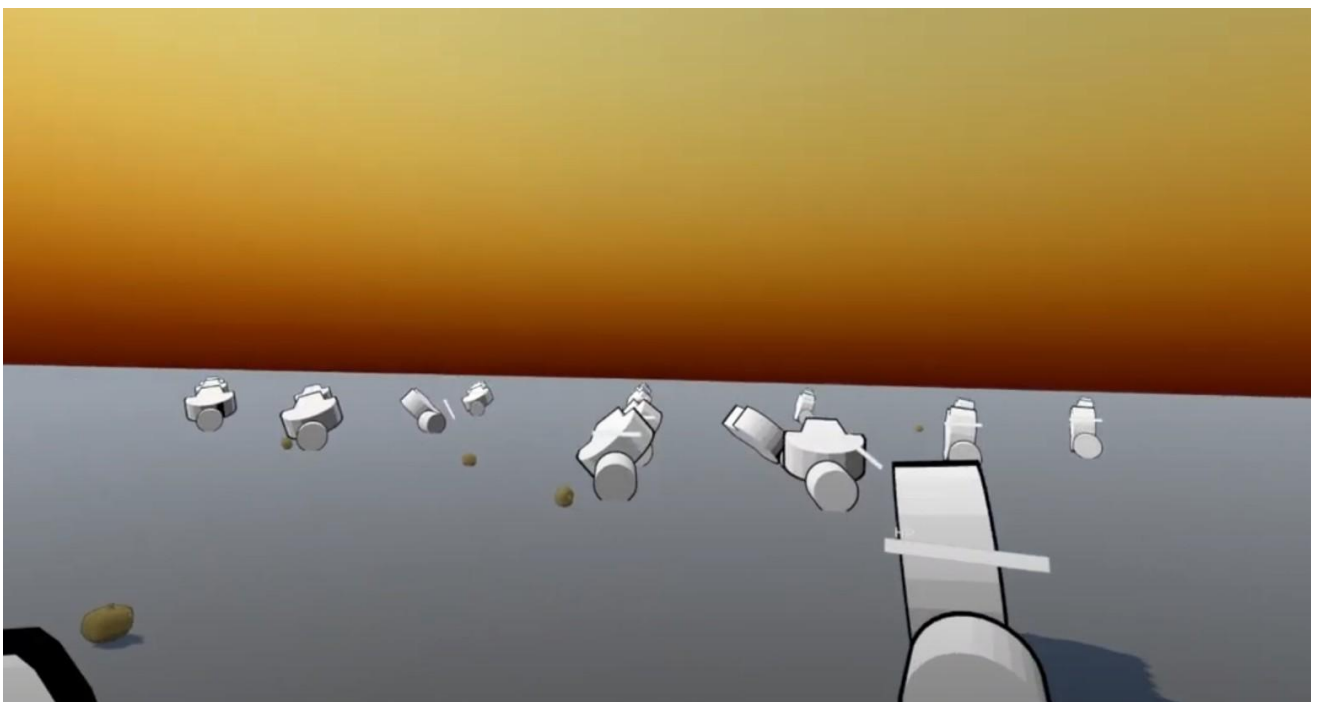
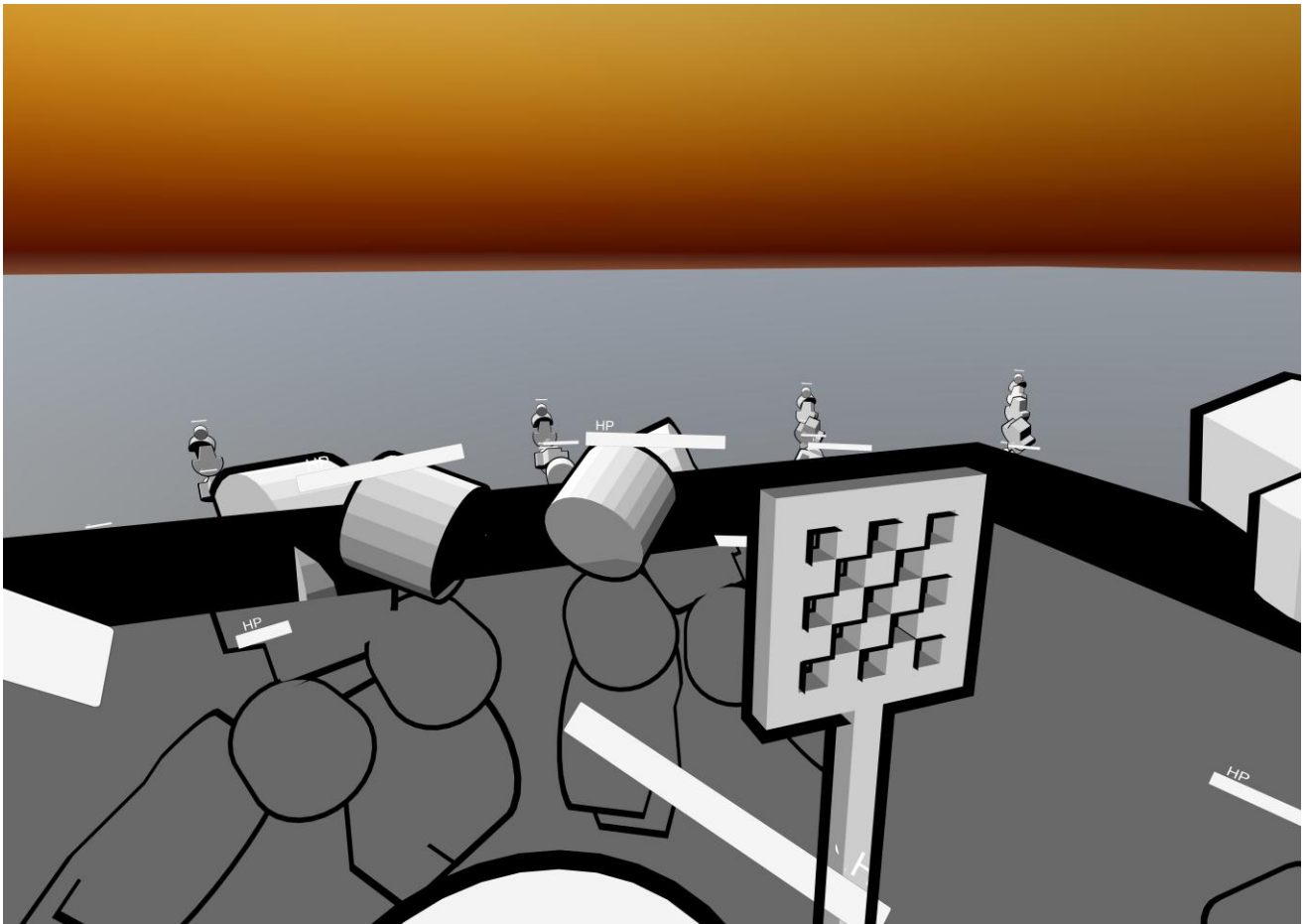


The supermarket environment was reconstructed from existing assets with:

- Low-poly self-created assets for shelves, props, and structural elements
- Modular prefab design for easy level expansion
- Realistic object scaling to enhance player immersion

Physics materials were tuned to provide appropriate friction and bounciness for different object types.

4.3.3 Enemy System



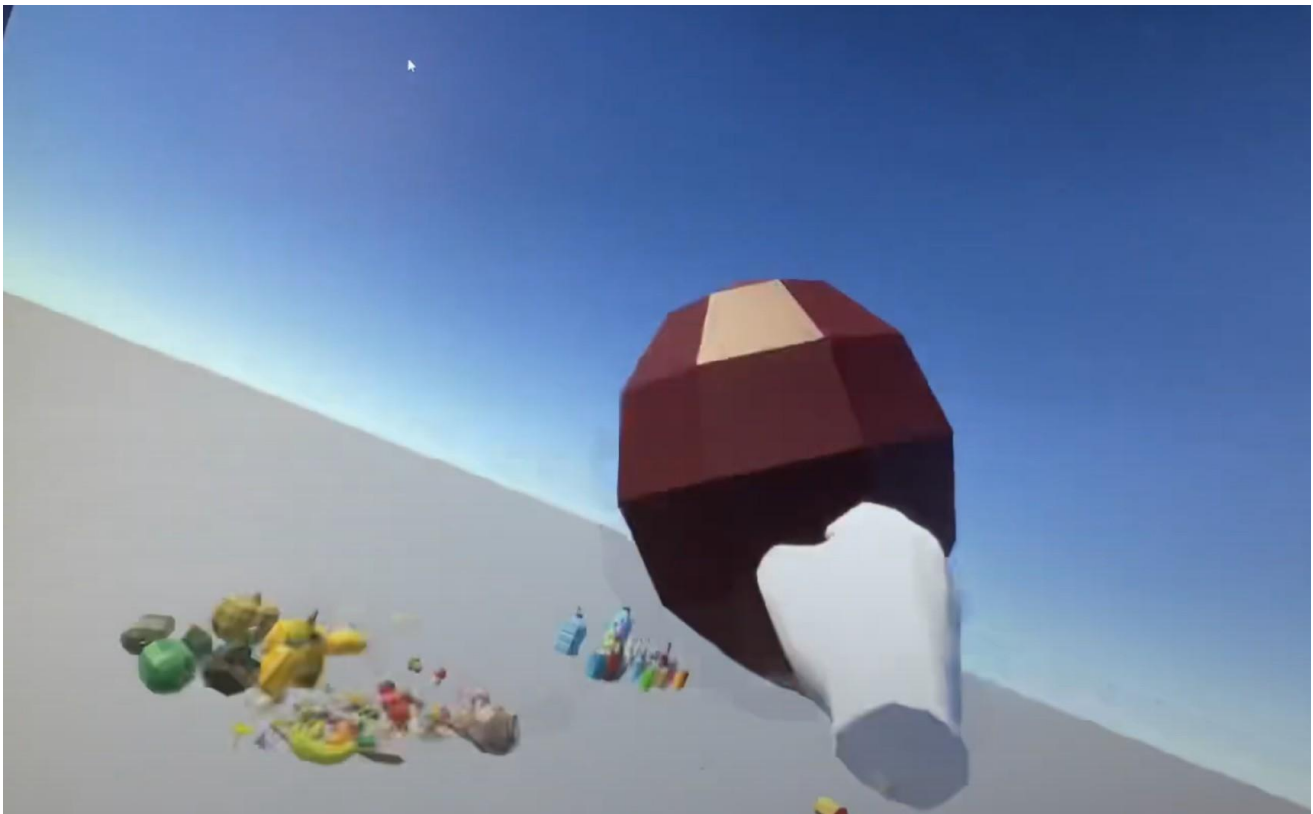
”Simple pursuit logic applied to enemy Rigidbody movement”

Enemies were implemented with a minimal but effective AI:

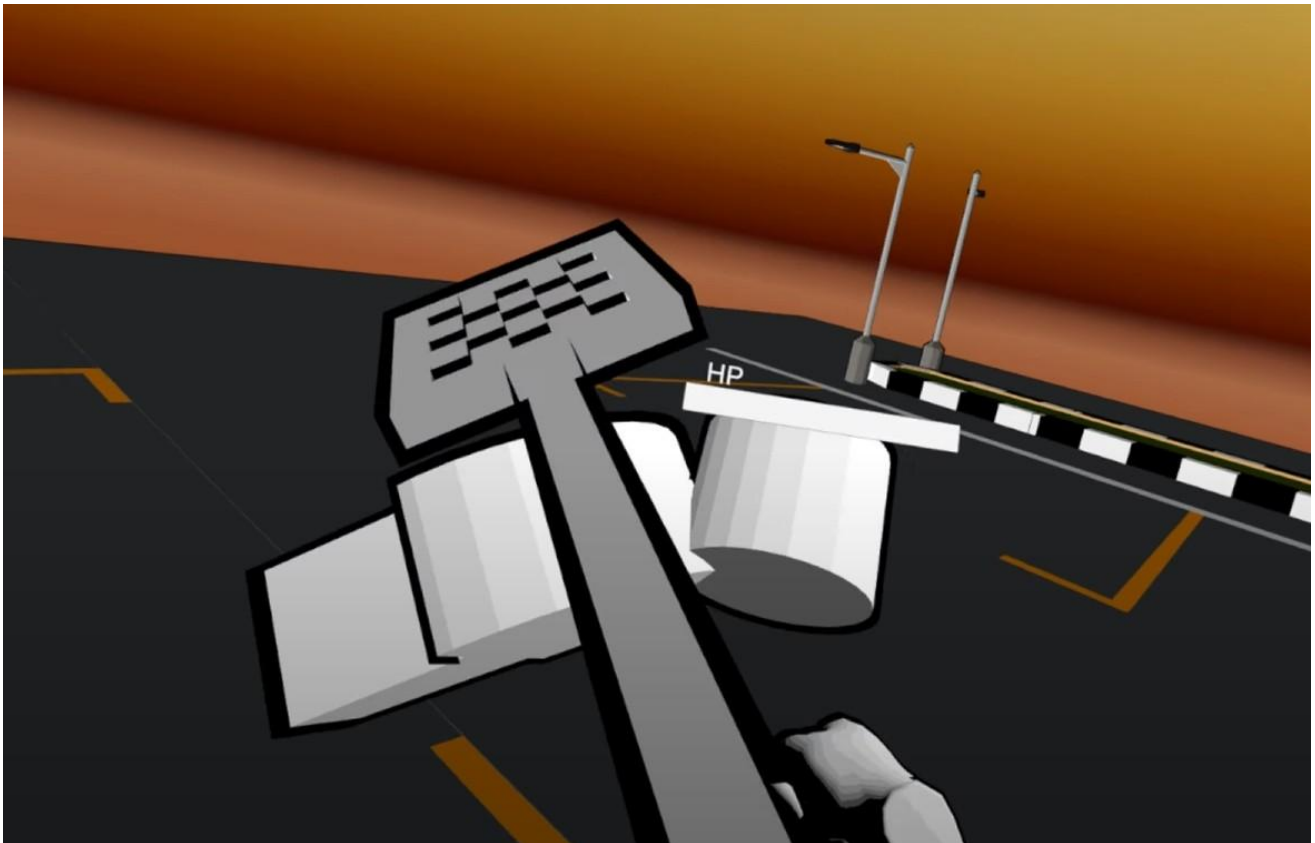
- A Rigidbody-based rolling movement system
- Head-tracking that dynamically faces the player
- Pursuit logic controlled by force acceleration rather than navmesh pathfinding because NavMesh would constrain emergent movement and feel too artificial for the intended rolling enemy behavior

Explosion and impact responses were handled entirely via Unity's built-in physics forces (AddExplosionForce and AddForceAtPosition), ensuring consistent, emergent reactions.

4.3.4 Interaction and Combat System



"Food item interaction testing using XR Grab Interactable system"



"Fly swatter melee weapon impact testing"

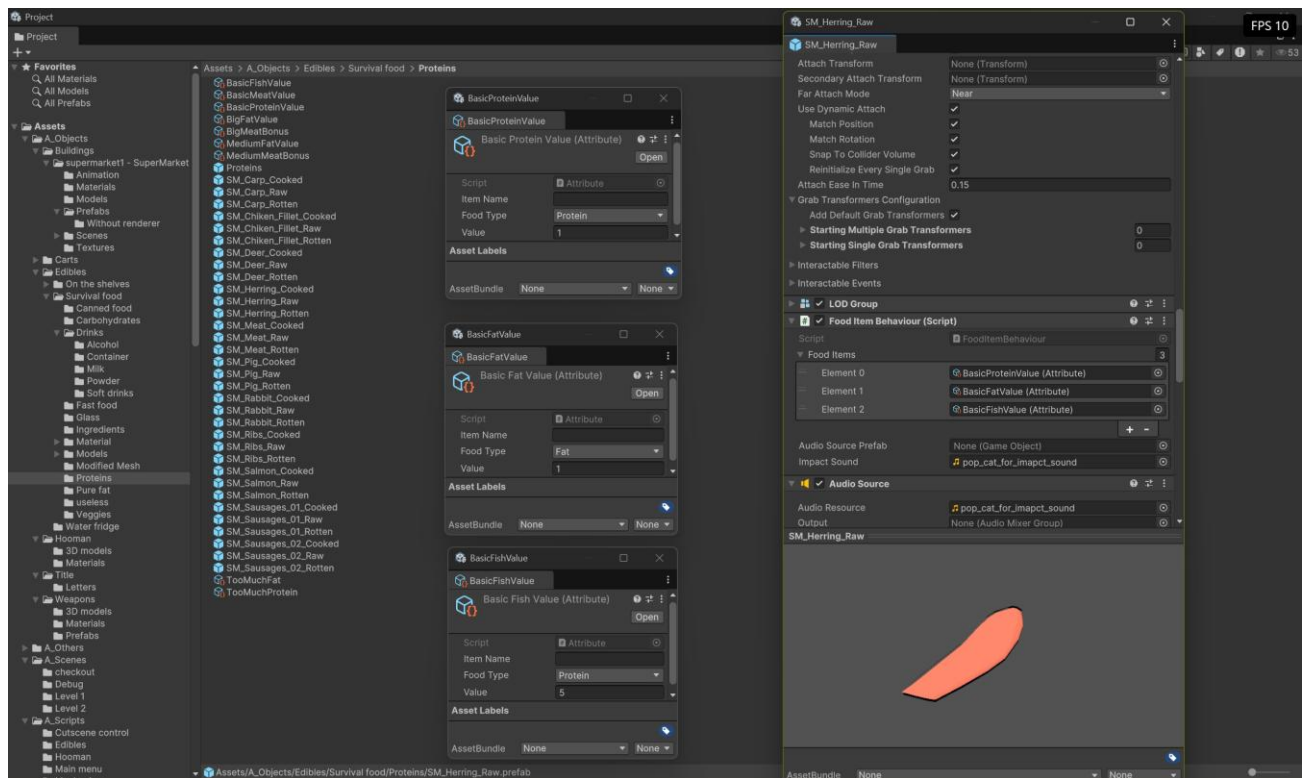
Every interactive object (weapons, food, props) was assigned:

- A Rigidbody
- Grabbable properties
- Optional special attributes (e.g., explosive on impact)

Melee combat was achieved by detecting collision force between player-held objects and enemy colliders, applying physics impulses accordingly.

No animation-driven damage systems were used, reinforcing the commitment to natural, emergent interaction.

4.3.5 Scoring System Design and Implementation



"Categorized food items for attribute assignment(raw herring)"



"Attribute-driven scoring flow in the trolley system"



"Final score"

Motivation

The scoring system was designed not simply as a mechanism for rewarding player actions, but as a subtle guidance tool to influence player behavior towards specific gameplay goals. Rather than awarding uniform points for all collected items, the system differentiated between food categories, encouraging players to prioritize fresh, unprocessed foods over canned or processed items.

This added a layer of strategic depth to an otherwise chaotic physical environment, subtly reinforcing themes of discernment and selection within the supermarket setting.

Technical Structure

The scoring system was implemented through two primary scripts: ExitChecking.cs and Scoring.cs.

Each food item in the environment was associated with a FoodItemBehaviour script containing an Attributes class instance that stored essential information, including food type, weight, and nutritional value indicators.

When a food item entered the Trolley collider trigger, ExitChecking.cs detected the event and retrieved the relevant attribute data.

The Scoring.cs script then calculated the points to award based on a predefined formula that weighted item categories differently, assigning higher scores to healthier food types and adjusting for weight.

Scoring updates were performed immediately upon item collection, providing real-time feedback to the player.

A live score display was attached above the Trolley within the player's field of view, allowing continuous tracking of their current performance.

Upon reaching the designated scoring zone (Exit Point), the final score was locked, evaluated, and presented through an end-of-session summary interface.

Challenges and Solutions

Several technical challenges emerged during the scoring system's development:

- **Duplicate Collision Events:**
Initially, food items that bounced within the trolley could trigger multiple scoring events.
This was mitigated by adding a verification flag to each item upon scoring, preventing repeated counts.
- **Attribute Data Consistency:**
Ensuring that all food items carried the correct and complete attribute data required meticulous prefab management and debugging tools to audit item setups.
- **UI Feedback Responsiveness:**
Score updates were optimized to occur with minimal frame delay, maintaining player immersion and ensuring that feedback felt immediate and satisfying.

Through iterative refinement, these challenges were addressed, resulting in a robust and engaging system that significantly enhanced the gameplay loop.

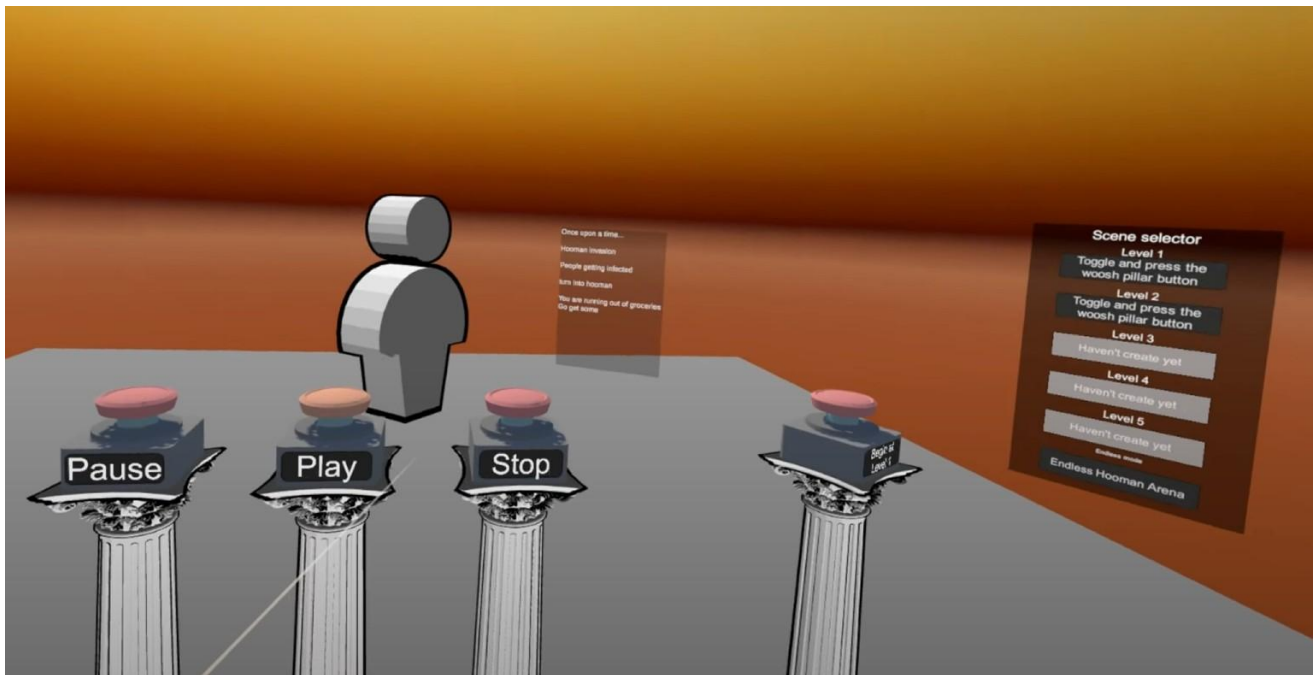
Impact on Gameplay Experience

The differentiated scoring approach introduced subtle but meaningful strategic layers into the game.

Players were naturally incentivized to move beyond random item collection, instead considering the types of food they selected and the physical challenges involved in transporting them under pressure.

This not only diversified player behavior but also reinforced the game's thematic focus on choice and resourcefulness, contributing to a richer and more fulfilling VR experience.

4.3.6 Cutscene Trigger System Design and Implementation



"Triggering system controlling cutscene sequences and level transitions"

Motivation

The Cutscene System was developed to enrich the game's narrative delivery without relying on conventional dialogue or external cinematic sequences.

Rather than building pre-rendered scenes, the approach focused on dynamically managing in-game objects, environmental states, and spatial transitions, preserving immersion while enhancing storytelling impact.

A modular control mechanism was required to flexibly manage different cutscenes and enable seamless level transitions based on player progression.

Technical Structure

The system was implemented primarily through the Triggering.cs script, which acted as a centralized controller for cutscene playback management.

Each cutscene was pre-assembled within the Unity environment as an independent Timeline or sequence, capable of being triggered, paused, resumed, or terminated via script.

Upon player interaction with specific trigger zones or event conditions, the system would execute predefined commands, such as:

- Playing a designated cutscene sequence
- Pausing an active cutscene (e.g., for dramatic timing or player interaction)
- Stopping a cutscene and transitioning to another gameplay phase
- Initiating a scene load to transition between game levels

This modular approach allowed maximum flexibility, enabling the integration of narrative beats without breaking the flow of the player's experience.

Challenges and Solutions

Key challenges in the development of the Cutscene System included:

- **Maintaining VR Comfort:**
Rapid visual changes could potentially cause motion sickness. Cutscene sequences were carefully constructed to minimize camera movements and avoid disorienting transitions.
- **Synchronizing Cutscenes and Level Transitions:**
Ensuring that cutscene completion naturally led into level loading without immersion breaks required tight coordination of event timing and resource unloading.
- **System Modularity:**
To future-proof the project for potential expansion, the cutscene control system was designed to allow easy addition of new sequences without rewriting core logic.

These challenges were addressed through systematic testing, conservative environmental manipulation, and the implementation of flexible, parameterized trigger methods.

Impact on Gameplay Experience

The Cutscene System added narrative weight and spatial dynamism to the supermarket environment.

Players were able to experience emotional shifts—such as heightened tension, relief, or surprise—through in-world events without needing conventional storytelling techniques.

By maintaining control within the immersive VR environment, the system preserved player agency while subtly guiding narrative pacing and world progression.

4.4 Testing Strategy

Testing followed a **build-play-observe-rebuild** cycle:

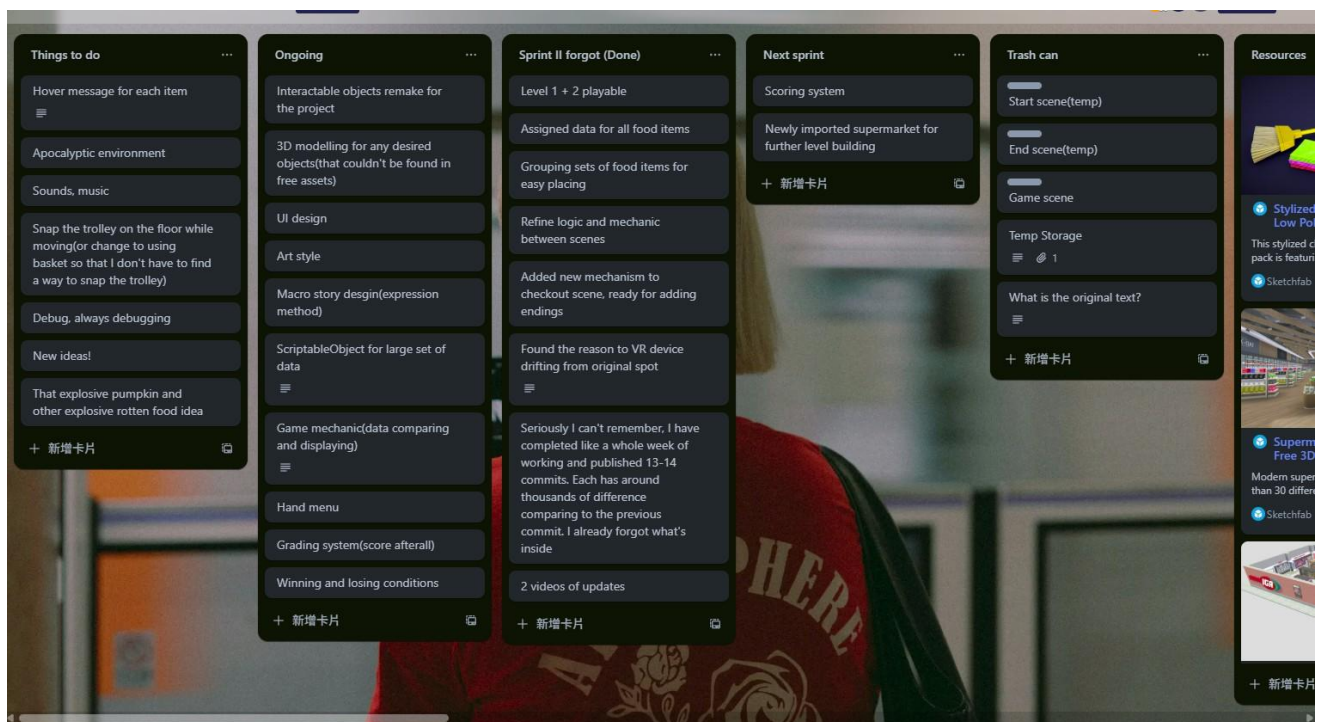
- After every major system implementation, in-VR testing sessions were conducted to evaluate physical feel, interaction reliability, and environmental coherence.
- Bugs and immersion breaks were logged immediately and addressed before proceeding.
- Informal peer testing was conducted periodically to validate assumptions about player behavior and emotional responses.

4.5 Project Management Approach

Effective project management was essential to ensure steady progression across all phases of development, particularly given the high interdependence between VR interaction systems, environmental design, and performance optimization.

Rather than adopting a traditional waterfall model, a **hybrid iterative approach** inspired by agile principles was employed to maintain flexibility and responsiveness.

4.5.1 Sprint-Based Development Cycle



"Sprint development cycle kept with Trello board used for weekly milestones"

	Fall into insanity new thereisnouse1 committed on Mar 21 00a5251
	Minor bugs fixed, creative opening update new thereisnouse1 committed on Mar 21 90b0d78
	Title settled thereisnouse1 committed on Mar 21 29a9278
⌵	Commits on Mar 20, 2025
	Initial sound effect update new thereisnouse1 committed on Mar 20 9a577e3
⌵	Commits on Mar 14, 2025
	Hooman Humidifier update new thereisnouse1 committed on Mar 14 bd8c612
⌵	Commits on Mar 13, 2025
	Minor bugs fixed new thereisnouse1 committed on Mar 13 f75739c
⌵	Commits on Feb 25, 2025
	Cutscene 2 completed, not tested yet thereisnouse1 committed on Feb 25 0f60f0a
⌵	Commits on Feb 22, 2025
	Hooman HP slider added thereisnouse1 committed on Feb 22 6fd2011
	An ending cutscene added and bugs fixed new thereisnouse1 committed on Feb 22 7857b02
⌵	Commits on Feb 21, 2025
	Indeed I am genius new thereisnouse1 committed on Feb 21 c5aef4a8

"Branching model in GitHub for version control and feature integration"

The project was divided into short, goal-focused sprints, typically spanning one to two weeks each.

Each sprint focused on delivering a specific functional increment—such as the XR grabbing system, the explosion mechanics, or environmental storytelling triggers—allowing for tangible progress without overcommitting to rigid long-term plans.

The sprint cycle typically followed this structure:

- **Sprint Planning:** Identification of the next set of achievable goals based on the current project state.
- **Sprint Execution:** Focused work on sprint tasks, allowing for emergent problem solving as technical challenges arose.
- **Playtesting and Review:** Immediate in-VR evaluation to assess physicality, performance, and player experience.
- **Sprint Retrospective:** Informal analysis of what succeeded, what needed adjustment, and how priorities should shift.

4.5.2 Risk-Responsive Adaptation

Flexibility was critical when unforeseen challenges emerged, such as:

- Physics instability after explosion force tuning
- Unexpected player behaviors during playtesting
- Performance drops during high-object-count scenes

Rather than rigidly adhering to original task orders, the management approach prioritized **responsive adaptation**:

- If a system under development revealed deeper technical problems (e.g., object collider jitter), subsequent tasks were reshuffled to focus on stabilizing the issue before proceeding.
- Playtesting feedback immediately informed task reprioritization, ensuring that player experience quality remained the central driving factor.

This nimble adaptation strategy ensured that the project could evolve organically without losing sight of its core goals.

4.5.3 Link to Development Timeline and Milestones

The sprint-based management approach was closely aligned with the broader development milestones outlined in Appendix A:

- Early sprints concentrated on establishing critical foundations (player movement, object interactions).
- Mid-project sprints expanded functionality (enemy AI behavior, grenade physics).
- Late-stage sprints emphasized polish (cutscene integration, sound layering, final performance tuning).

This structured yet adaptable methodology allowed the project to reach functional completion while preserving creative flexibility, a necessity given the experimental nature of the game's design philosophy.

Chapter 5: Legal, Social, Ethical and Professional Issues (LSEP)

5.1 Legal Considerations

5.1.1 Intellectual Property and Licensing

Throughout development, strict adherence to intellectual property (IP) laws was maintained:

- All models, textures, and scripts were either **self-created**, sourced from **royalty-free repositories**, purchasing, or adapted under **Creative Commons licenses** with appropriate attribution.
- No copyrighted third-party materials were used without explicit permission.
- Sound effects used in the game were generated using AI tools.

This ensured the project remained legally compliant, mitigating risks of infringement if published or distributed beyond academic purposes.

5.1.2 Platform Compliance

Although the project remained an academic prototype, Oculus platform guidelines were referenced to ensure future compatibility, particularly regarding:

- Prohibitions on excessive graphical violence
- Safety warnings for VR gameplay
- Technical optimization standards

By following these guidelines early, the project would require minimal adjustment for potential public release.

5.2 Social Considerations

5.2.1 Accessibility

VR development inherently faces accessibility challenges, including physical space requirements, mobility limitations, and sensory impairments.

While full accessibility support was beyond the project's scope, the following considerations were made:

- The game was designed to be playable **entirely standing still**, minimizing the need for large physical play spaces.
- No auditory cues were made critical for progression, respecting players with hearing impairments.

Future iterations would aim to incorporate seated play options and customizable interaction methods.

5.2.2 Psychological Impact

Despite involving confrontation with humanoid enemies, the game was deliberately designed to:

- Avoid realistic gore, dismemberment, or graphic violence.
- Utilize abstracted, low-polygon models for enemies to maintain emotional distance.
- Allow players to disengage from conflict entirely, reinforcing autonomy rather than aggression.

This approach minimized risks of triggering trauma responses or causing distress in vulnerable players.

5.3 Ethical Considerations

5.3.1 Player Safety in VR

Virtual reality poses unique physical risks, including:

- Motion sickness
- Physical collisions with real-world obstacles
- Eye strain and disorientation

To mitigate these risks:

- No artificial locomotion (e.g., joystick movement) was implemented; player movement corresponded directly to real-world motion.
- Cutscenes used **fade transitions** instead of camera jolts to avoid disorientation.

5.3.2 Depiction of Violence

Ethical standards for VR differ from flat-screen games because immersion amplifies emotional reactions.

The project consciously adopted a non-graphic, almost absurdist depiction of violence:

- Enemies collapse or tumble rather than displaying pain or suffering.
- No blood, dismemberment, or explicit death animations are presented.
- Players can experience, interpret, or avoid conflict without narrative punishment.

Thus, the project aligns with ethical best practices, avoiding glorification of violence while still enabling physical interaction.

5.4 Professional Conduct

Professional standards were upheld throughout development:

- **Version Control:** Local iterative backups were maintained at every milestone.
- **Feedback Integration:** Player feedback, even informal, was collected and evaluated for improvement opportunities.
- **Testing Under Realistic Conditions:** All builds were tested on real hardware (Oculus Quest 2) rather than relying on simulated environments.

Moreover, the project maintained a commitment to **honest representation**:

- No exaggerated claims were made regarding functionality.
- All known limitations were acknowledged and documented transparently within the report.

This adherence to professional ethics demonstrates readiness for future industry or research participation.

Chapter 6: Project Management

6.1 Project Planning

Given the solo nature of the project, a **lightweight agile methodology** was adopted, drawing inspiration from Scrum but tailored to the flexibility and unpredictability of individual work.

Rather than strict sprint schedules or formal daily stand-ups, a **milestone-based roadmap** was constructed at the start of development, identifying critical deliverables and logical phases of progression.

The project was divided into four major phases:

Steps	Objective
Step 1: Foundation Setup	Establish XR Rig, base player movement, and object interaction system.
Step 2: Core Systems Development	Implement enemy AI behavior, object attributes, basic environment construction, scoring system development for to publish the Minimum Viable Product(MVP).
Step 3: Interaction Refinement	Focus on physical feedback, explosion mechanics, environmental storytelling systems (cutscenes).
Step 4: Final Polish and Testing	Bug fixing, visual enhancements, sound layering, usability improvements, documentation writing.

Each step was timeboxed with approximate weekly goals, while remaining flexible to accommodate testing feedback and emergent design challenges.

6.2 Task Breakdown and Tracking

Tasks were broken down using a handwritten, simple sprint board approach:

- **Backlog:** List of all identified tasks, both critical and optional.
- **In Progress:** Tasks actively being worked on during each mini-sprint.
- **Done:** Completed and tested features.

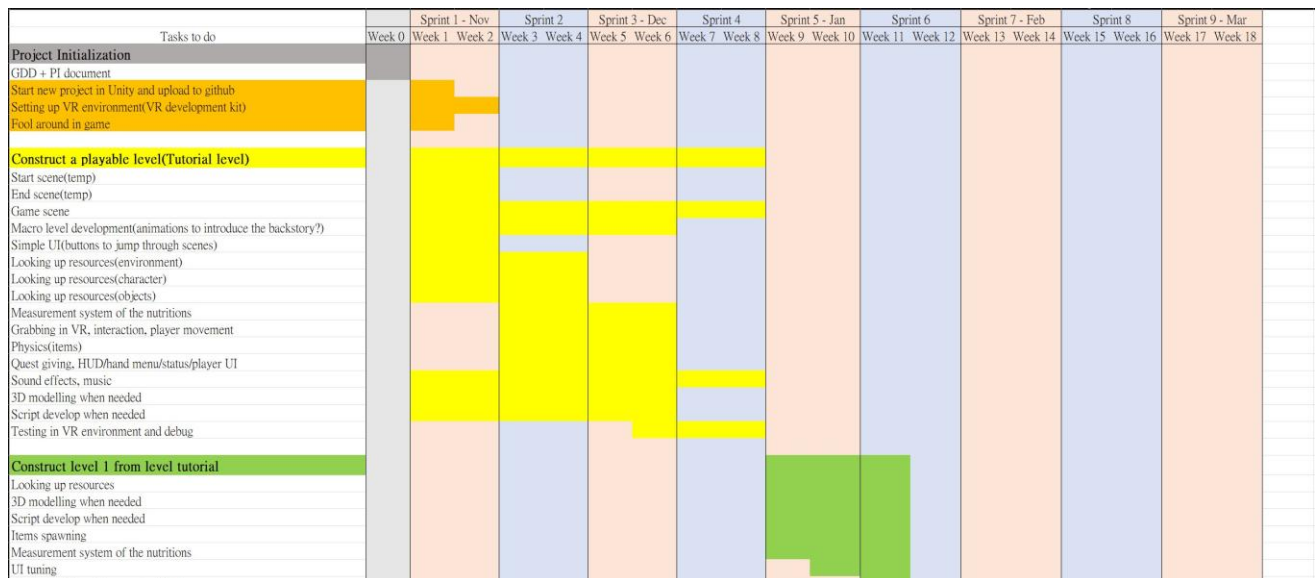
This lightweight tracking method allowed rapid reprioritization when issues arose (e.g., shifting focus to explosion tuning after initial player feedback highlighted the weakness of impact feel).

Key tracked categories included:

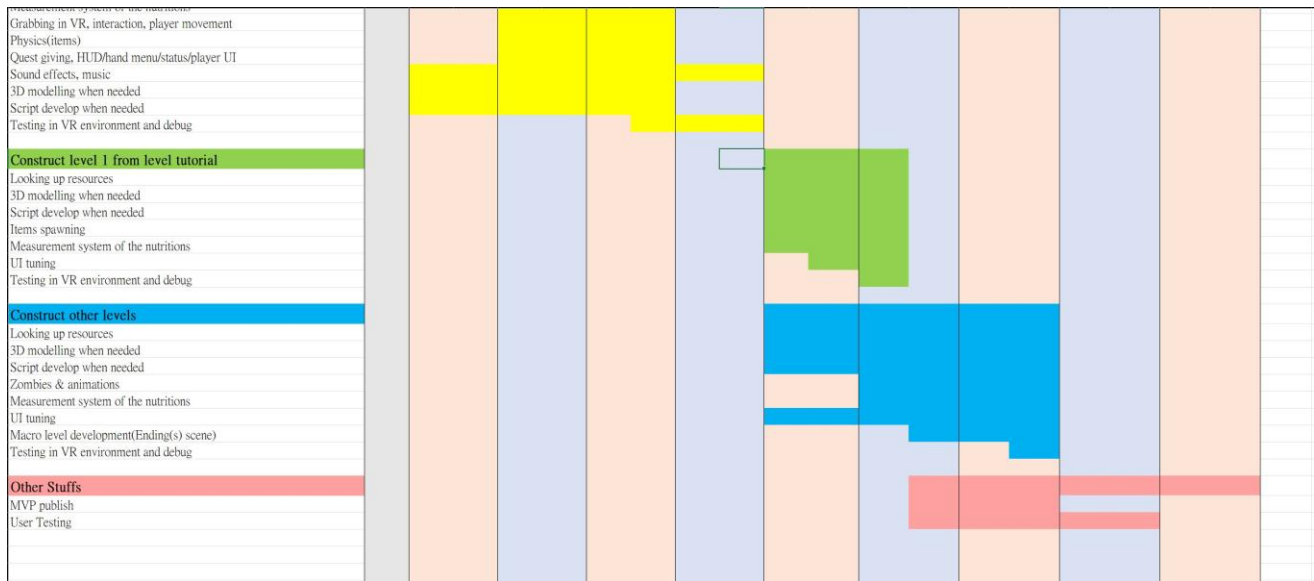
- Player interaction systems
- Enemy movement and response systems
- Environment construction
- Cutscene scripting and trigger placement
- Audio integration
- Optimization passes

An example sprint board outline is included in the appendices.

6.3 Gantt Chart Overview



"Gantt chart part 1"



"Gantt chart part 2"

Deliberate time buffers were included toward the end of the timeline to account for unforeseen technical issues and last-minute polish. Flexible time arrangement is set for each sprint within 1-2 weeks in latter project management.

6.4 Risk Management

Recognizing the inherent risks in VR development and physics-based systems, a basic risk assessment strategy was maintained throughout:

Risk	Mitigation Strategy
Player motion sickness	No artificial movement; fade transitions during cutscenes.
Physics instability (object clipping, jitter)	Extensive collider and Rigidbody tuning; friction materials applied.
Performance drops	Object pooling and grouping, optimized rigidbody counts, low-poly models.
Scope creep	Strict adherence to core interaction goals; avoidance of unnecessary feature additions.

Risk logs and problem-resolution notes were kept during major milestones and are summarized in the appendices.

6.5 Reflection on Management Process

While lightweight, the adopted project management structure proved highly effective:

- **Flexibility** allowed for creative experimentation without derailing timelines.
- **Continuous playtesting and micro-revisions** ensured that feature quality was prioritized over raw feature quantity.
- **Risk awareness** prevented catastrophic late-stage technical failures.

In future projects, even tighter formal tracking (such as automated sprint tracking tools) could be beneficial, but for the scope and nature of this project, the management approach was appropriately scaled and successfully executed.

6.6 Version Control

In order to maintain a clear history of changes, support experimentation, and recover quickly from errors, two complementary tools were used throughout development:

GitHub

A single GitHub repository was created at the outset. Two primary branches were maintained at all times:

1. **master** – holds the latest stable build, suitable for in-VR testing or submission at any time
2. **develop** (or **temp**) – used as the integration branch for ongoing feature work

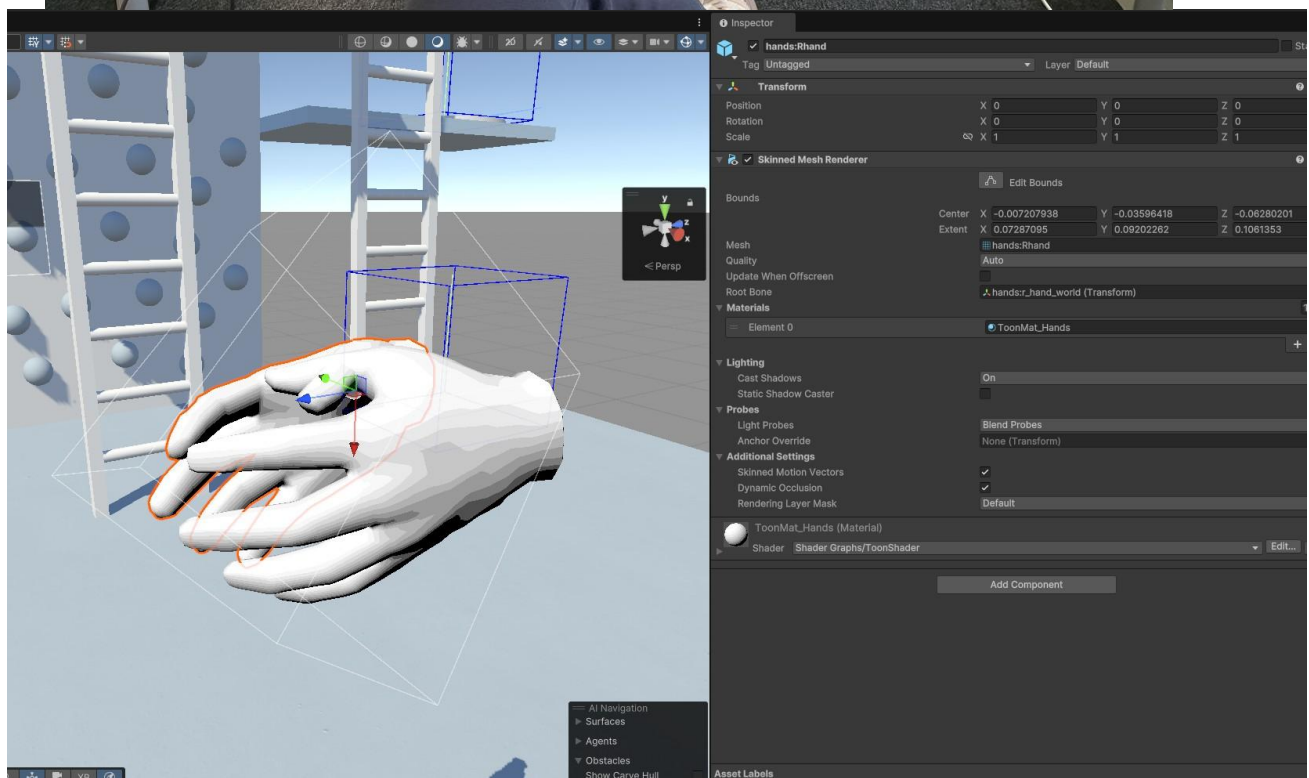
Trello for Task and Version Control Coordination

- **Board Organization**

A Trello board supplemented GitHub by visualizing planned, in-progress, and completed tasks. Each card represented one discrete piece of work

6.7 Sprint Development insight

Sprint 1: Foundation Setup



”Configuration of XR Rig prefab structure and hand anchor calibration for initial VR setup.”

Overview

The first sprint of the project was dedicated to establishing the basic XR environment necessary for VR gameplay.

The primary focus was setting up the XR Origin Rig within Unity and ensuring correct hand tracking and camera behavior for Oculus Quest 2.

Challenges encountered during this phase included initial calibration errors causing minor camera drift and misalignment between controller inputs and the visual representation of hands.

These foundational issues were systematically identified through iterative in-VR testing and promptly resolved, laying a stable technical base for all future development.

Securing a functional, comfortable player presence was considered essential before proceeding to object interaction or environmental design.

Objectives

- Integrate XR Interaction Toolkit and Oculus SDK into the Unity project
- Build and configure a custom XR Origin Rig suitable for Oculus Quest 2
- Validate and correct hand tracking, controller input, and headset positioning
- Conduct initial standing-play calibration tests

Implementation Methods

The XR Rig was established using Unity's XR Interaction Toolkit, customized to match the Oculus Quest 2 hardware profile.

Hand anchors and colliders were resized for natural object grabbing.

Continuous in-VR validation helped fine-tune player height offset and controller mappings, ensuring minimal positional drift and establishing a comfortable standing experience.

Sprint 2: Core Interaction Prototyping”Looking for trolley asset”



“Hand grabbing testing”



“Testing grab-release dynamics and throw physics with varied object weights.”

Overview

Building upon the XR Rig foundation established in sprint 1, the second sprint concentrated on implementing basic object interaction mechanics.

The primary aim was to enable players to naturally grab, hold, and throw virtual objects within the supermarket environment.

Significant attention was given to the physicality of interactions, ensuring that object grabbing felt responsive and intuitive.

Early tests revealed inconsistencies in grip stability and object release trajectories, particularly when players moved their hands quickly.

To address these issues, adjustments were made to Rigidbody mass values, collider shapes, and grab detection thresholds.

This sprint marked the beginning of physical engagement with the virtual world, moving beyond passive exploration towards active manipulation.

Objectives

- Implement basic grab and release mechanics for interactable objects
- Fine-tune Rigidbody settings to ensure realistic object weight and behavior
- Validate hand pose alignment during grabbing and releasing
- Test preliminary object throwing behavior

Implementation Methods

Using the XR Grab Interactable system, a base script for object interaction was created and customized for supermarket props.

Rigidbody mass values were adjusted according to object type, improving realism during manipulation.

Throwing trajectories were tested extensively, leading to minor collider and grip detection tweaks to ensure responsive and intuitive object releases.

Sprint 3: Basic Enemy Behavior Prototype



"Early development of Rigidbody-based enemy pursuit and tracking behavior."

Overview

Following the establishment of object interaction, development shifted focus toward creating dynamic threats within the supermarket environment.

Sprint 3 was primarily dedicated to prototyping the "Hooman" enemy behavior—designing a non-humanoid, rolling entities capable of pursuing the player while maintaining physical plausibility.

The initial goal was to implement basic autonomous movement toward the player without complex navigation systems like NavMesh, which would not align with the rolling movement style.

Early prototypes demonstrated that force-based movement provided a more organic feel, but balancing pursuit speed and preventing physics instability (such as flying or jittering) proved challenging.

This sprint established the mechanical basis for emergent tension through environmental encounters rather than scripted conflict.

Objectives

- Design and implement basic enemy (Hooman) prefab structure
- Develop a pursuit system using Rigidbody forces instead of NavMesh
- Implement dynamic head tracking to face player position
- Tune acceleration and maximum movement speed to simulate natural pursuit

Implementation Methods

A simple Rigidbody-driven AI script was attached to the Hooman prefab, applying forward forces based on the vector toward the player's position.

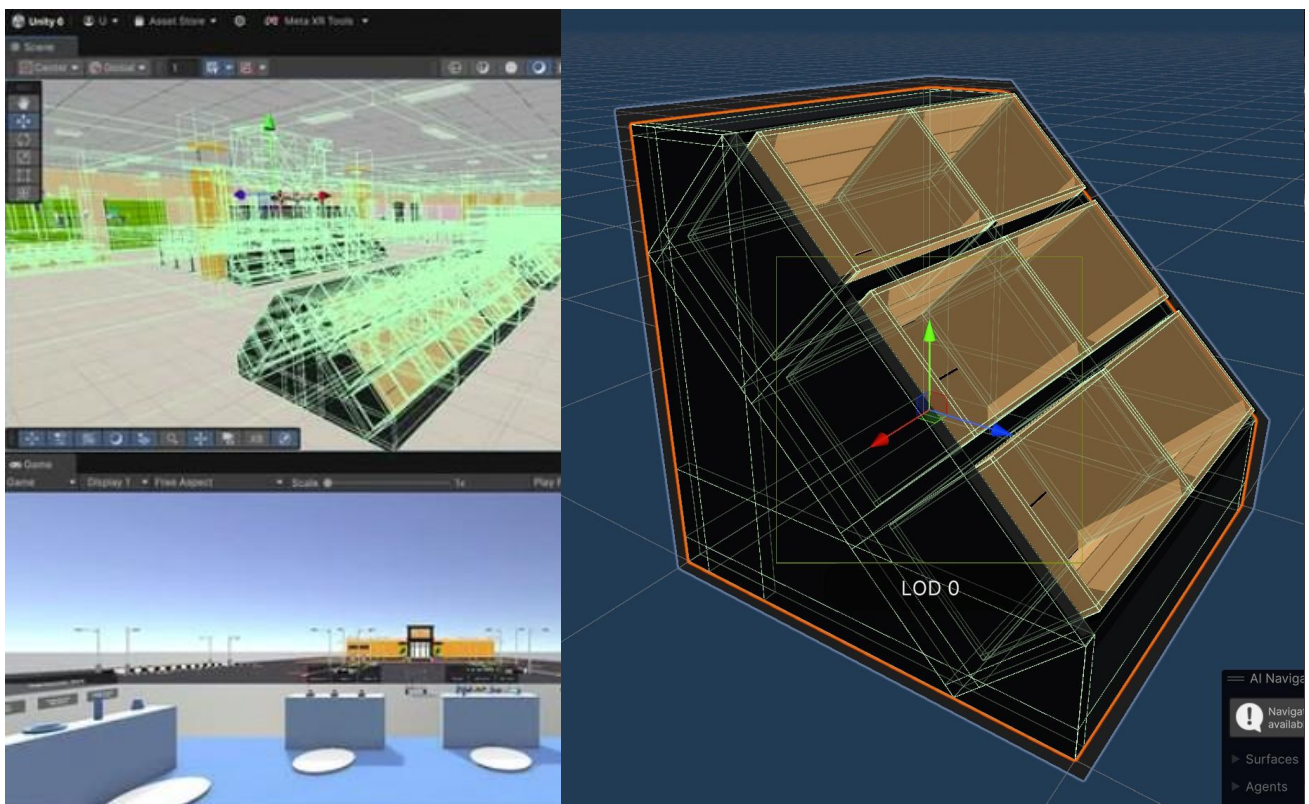
A separate head tracking system used Quaternion interpolation to smoothly rotate the Hooman's head toward the player each frame.

Speed and acceleration were exposed as tunable variables, allowing for quick iteration during playtesting to balance pursuit pressure and maintain physics stability.

Sprint 4: Supermarket Environment Construction



"Construction of modular supermarket environment using low-poly modular assets."



"Remake simple effective colliders for the assets"

Overview

With the core interaction and enemy systems functioning at a basic level, attention shifted toward building the supermarket environment that would serve as the game's primary setting.

The goal for sprint 4 was to create a convincing, navigable space that balanced realism with the project's low-poly cartoon aesthetic.

This involved constructing modular shelf units, checkout counters, and structural elements such as walls and ceilings.

A particular challenge during this phase was maintaining object scale and spacing to ensure comfortable navigation within VR, avoiding scenarios where shelves were either too cramped or too sparse.

In parallel, colliders were strategically placed and optimized to avoid performance degradation while preserving physical fidelity.

This sprint laid the physical foundation for emergent gameplay by providing a dense, interactive environment filled with tangible props.

Objectives

- Model and assemble modular supermarket components (shelves, counters, walls)
- Establish consistent real-world scaling across environment assets
- Set up colliders for major structures to ensure physics integrity
- Begin populating environment with preliminary props for interaction testing

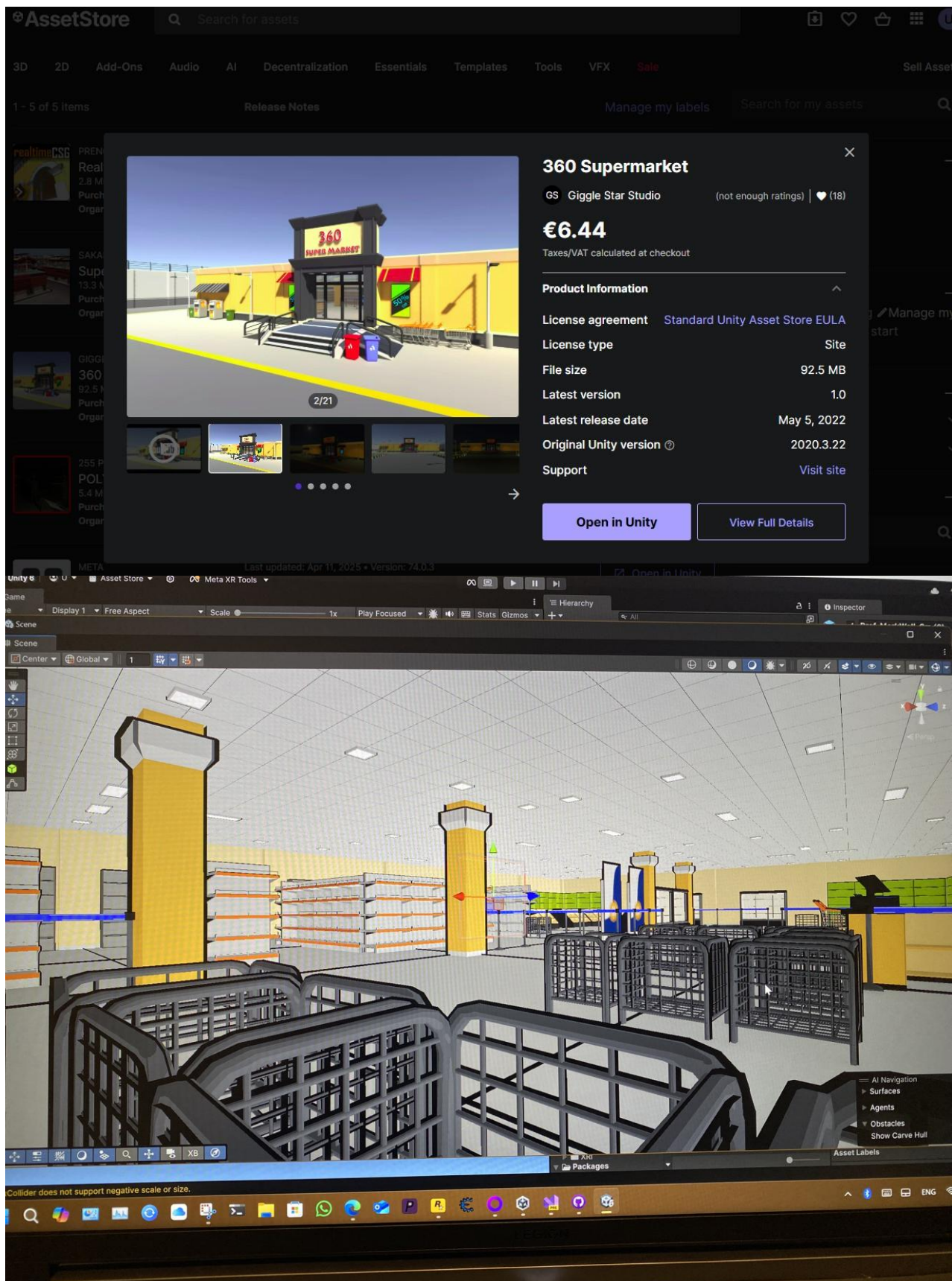
Implementation Methods

Modular environment assets were recreated using low-polygon modelling techniques, ensuring that each unit (e.g., shelf) could be easily duplicated and arranged.

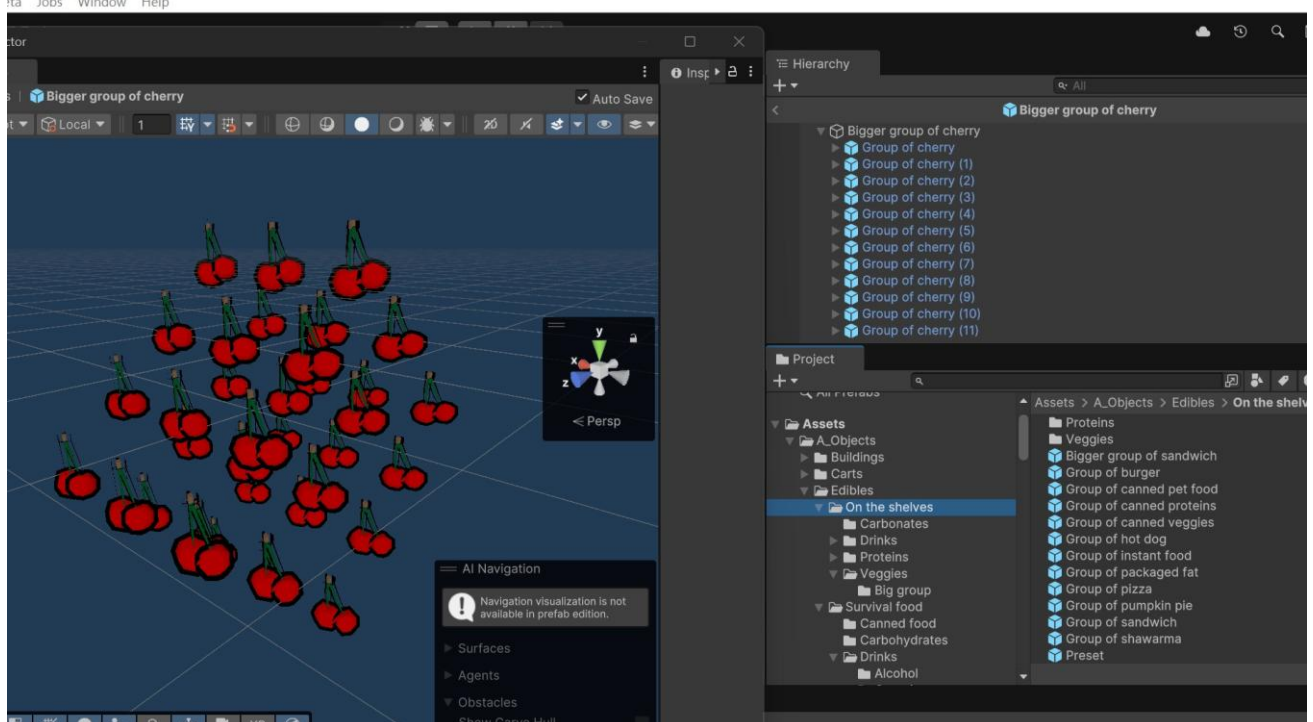
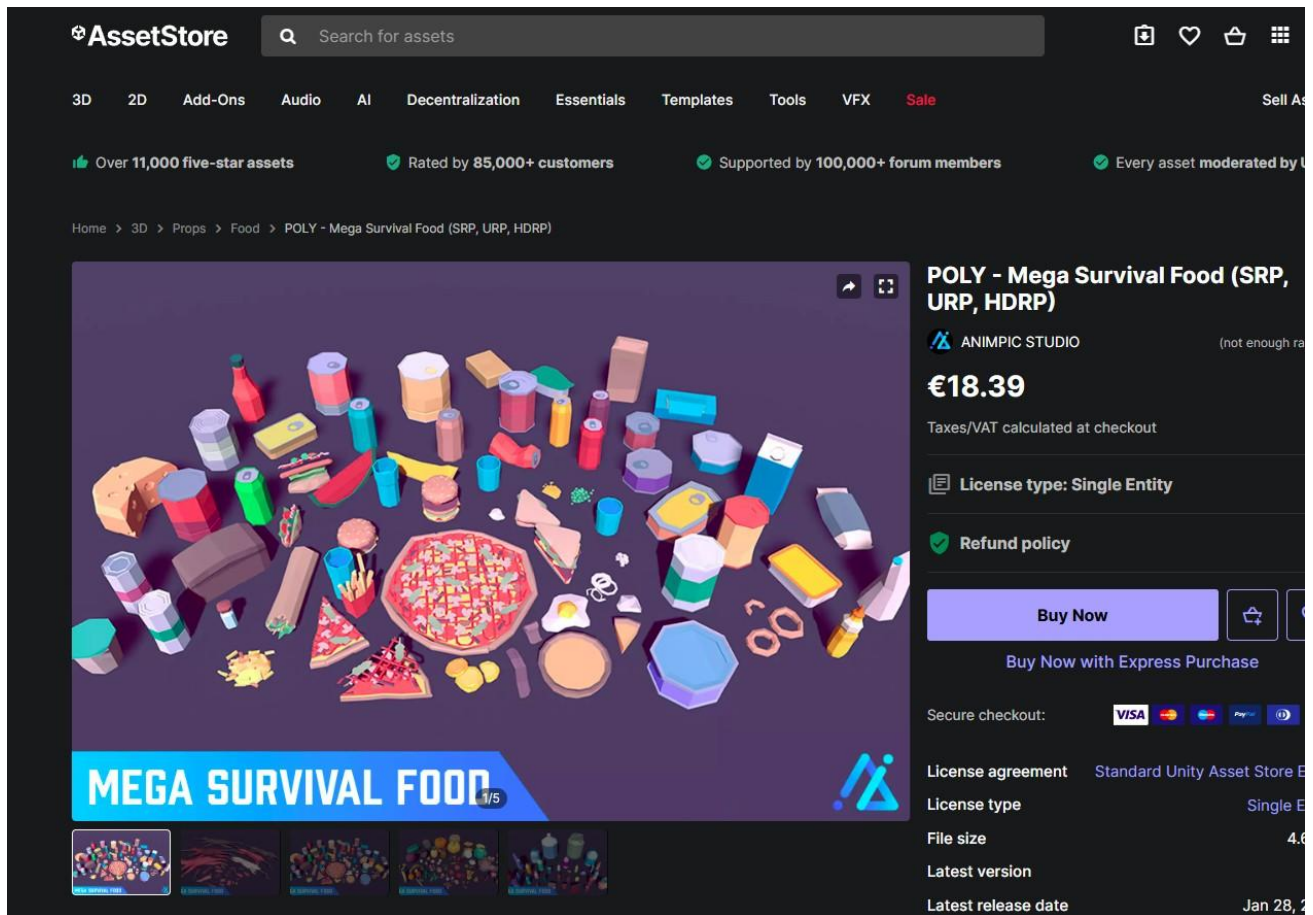
All assets were scaled relative to average human height for comfortable VR navigation. Box colliders and mesh colliders were selectively applied based on object complexity to optimize performance without sacrificing interaction quality.

Test builds were loaded onto the Oculus Quest 2 to validate real-world spatial coherence and ensure frame rates remained stable within the supermarket setting.

Sprint 5: Environment Asset Rework and Food Item Categorization



“Visual comparison between pre-existing assets and redesigned supermarket models”



“Visual comparison between pre-existing assets and redesigned models”

Overview

During sprint 5, the focus shifted toward refining the visual and structural consistency of the supermarket environment.

It became apparent that some of the downloaded assets used for shelves, counters, and food props did not align aesthetically with the project's emerging low-polygon, cartoon-inspired visual style.

To address this, existing environment models were reworked or entirely replaced with custom-designed assets that matched the intended stylistic coherence.

Simultaneously, all food item models were systematically categorized into several groups (e.g., fruits, veggies, beverages, proteins, drinks) to facilitate future attribute(s) assignments and gameplay mechanics.

This restructuring was essential for maintaining immersion, enabling consistent interaction behavior, and preparing the groundwork for dynamic systems like the trolley scanner logic.

Objectives

- Review and rework supermarket structural assets to align with cartoon-style visual direction
- Redesign or simplify complex downloaded models where necessary
- Systematically categorize all food item models into logical groups
- Prepare asset tagging for future attribute-based gameplay mechanics

Implementation Methods

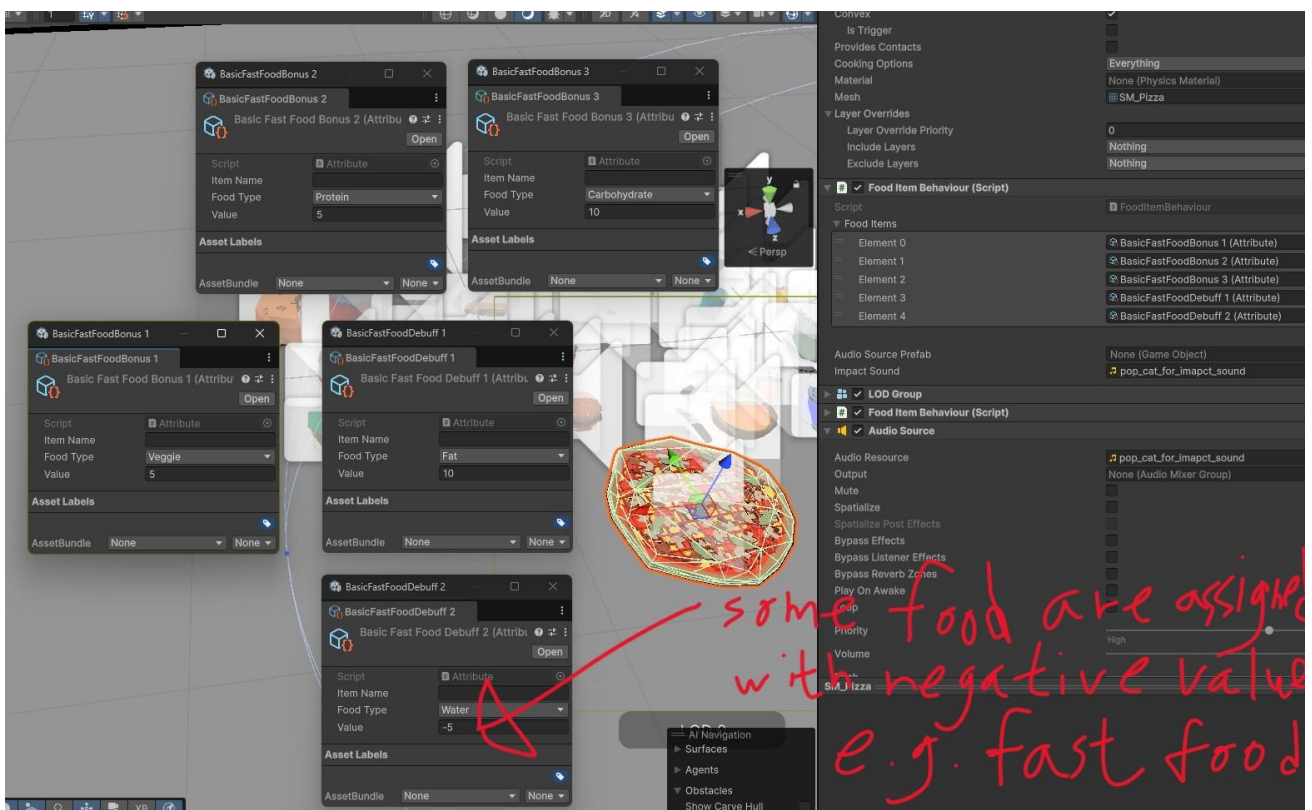
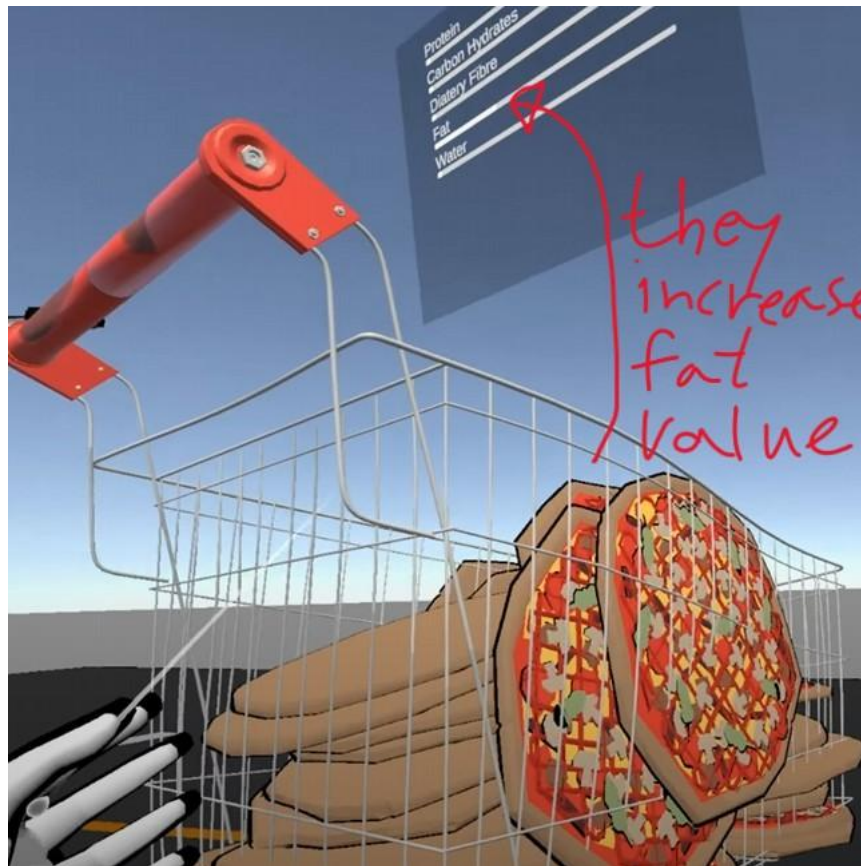
Custom low-polygon versions of key supermarket structures were modelled and integrated into the environment, replacing aesthetically inconsistent assets.

Material settings were adjusted to remove unnecessary reflections and normal maps, reinforcing the cartoon style.

Food item models were reviewed individually and grouped based on real-world similarities, with placeholder scripts prepared for future attribute tagging.

Organizational consistency was maintained across the Unity project hierarchy to streamline future scripting and scene management.

Sprint 6: Food Item Attribute System and Trolley Interaction



```

Unity Script (2 asset references) | 1 reference | thereisnouseername1, 66 days ago | 1 author, 3 changes
public class FoodManager : MonoBehaviour
{
    public Slider veggieSlider;
    public Slider proteinSlider;
    public Slider carbonSlider;
    public Slider fatSlider;
    public Slider waterSlider;

    private float veggieAmount;
    private float proteinAmount;
    private float carbonAmount;
    private float fatAmount;
    private float waterAmount;

    // Method to add food item
    1 reference | thereisnouseername1, 66 days ago | 1 author, 1 change
    public void CollectFood(Attribute food)
    {
        switch (food.foodType)
        {
            case FoodType.Veggie:
                veggieAmount += food.value;
                veggieSlider.value = veggieAmount;
                break;
            case FoodType.Protein:
                proteinAmount += food.value;
                proteinSlider.value = proteinAmount;
                break;
            case FoodType.Carbohydrate:
                carbonAmount += food.value;
                carbonSlider.value = carbonAmount;
                break;
            case FoodType.Fat:
                fatAmount += food.value;
                fatSlider.value = fatAmount;
                break;
            case FoodType.Water:
                waterAmount += food.value;
                waterSlider.value = waterAmount;
                break;
        }
    }
}

```

store value to slider

```

public class Trolley : MonoBehaviour
{
    public FoodManager foodManager;
    List<Attribute> foodItems;

    Unity Message | 0 references | thereisnouseername1, 66 days ago | 1 author, 5 changes
    private void OnTriggerEnter(Collider other)
    {
        // buggy, it could treat player as children game object
        //if (!other.CompareTag("Floor"))
        //    try to fix the physics lag
        //    other.transform.SetParent(this.transform);
        // indeed it fixed, thanks chatgpt

        // Check if the object entering is a food item
        // Attributes foodItem = other.GetComponent<FoodItemBehaviour>()?.foodItem;

        //bool value = other.TryGetComponent<FoodItemBehaviour>(out FoodItemBehaviour foodItem);

        //if(other.GetComponent<FoodItemBehaviour>() != null)

        //foodItem = other.GetComponent<FoodItemBehaviour>()?.foodItem;
        foodItems = other.GetComponent<FoodItemBehaviour>()?.foodItems;

        if (foodItems != null)
        {
            foreach (Attribute foodItem in foodItems)
            {
                // Add food item to the food manager
                foodManager.CollectFood(foodItem);
            }
            // Destroy(other.gameObject);

            /// moved to outside
            // try to fix the physics lag
            other.transform.SetParent(this.transform);
            // indeed it fixed, thanks chatgpt
            ///
        }
    }

    Unity Message | 0 references | thereisnouseername1, 66 days ago | 1 author, 5 changes
    private void OnTriggerExit(Collider other)
    {
    }
}

```

call function

“Dynamic food attributes system driving score calculation based on player actions”

Overview

Following the asset rework, sprint 6 focused on enhancing the gameplay depth by implementing a structured attribute system for food items.

The primary goal was to enable each food object to carry distinct data properties, such as weight, type, nutritional value, or freshness, which could later be used for scoring and interaction purposes.

In parallel, work began on the Trolley system, which would detect when food items entered its collider zone and retrieve their attributes dynamically.

Early prototype tests confirmed that linking Rigidbody food objects to an attribute system was feasible, but challenges emerged around ensuring reliable detection without physics glitches, especially when multiple items entered the trolley simultaneously.

By the end of the sprint, a basic framework was in place for dynamic item evaluation and score calculation based on item properties.

Objectives

- Design and implement a modular Food Item attribute system
- Attach attribute data to all existing food models
- Create a Trolley system capable of detecting food item entry
- Begin integrating basic scoring logic based on collected item attributes

Implementation Methods

A FoodItemBehaviour script was created and improved, encapsulating a serializable Attributes class containing food-specific properties.

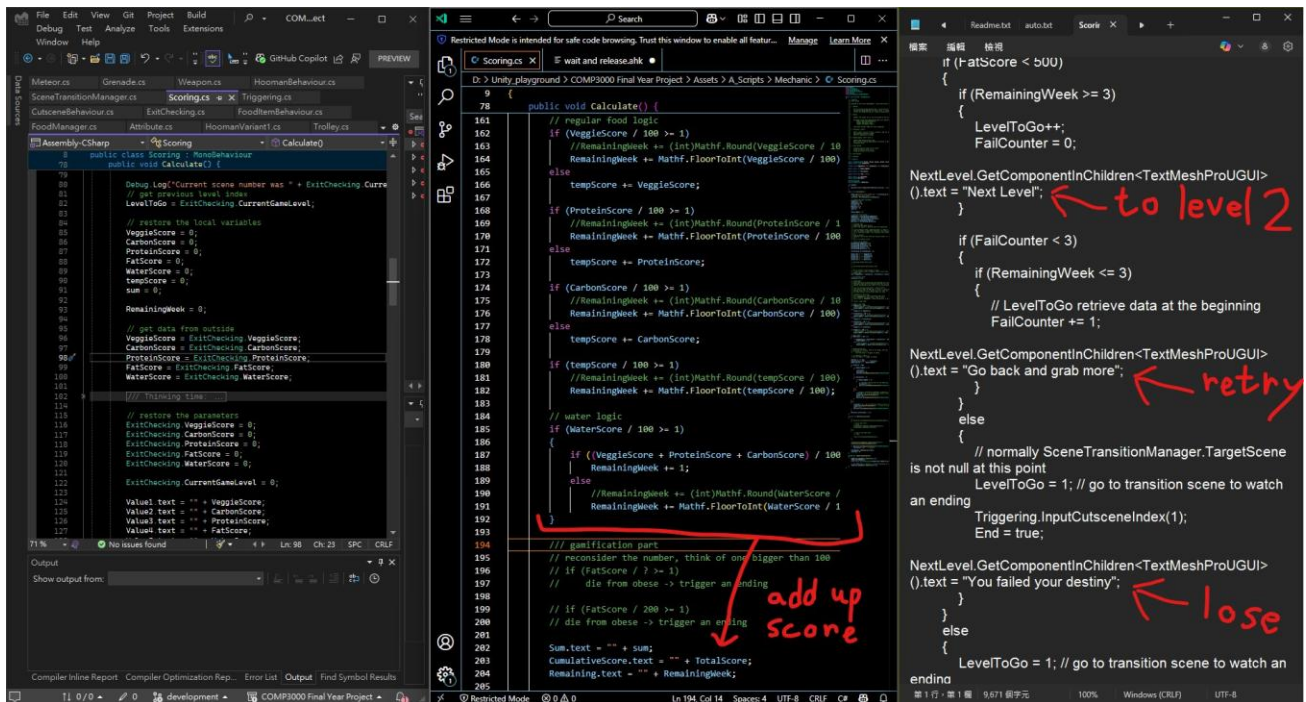
Each food prefab was assigned individualized attribute values.

The trolley collider was configured as a trigger zone, detecting food entry and accessing attribute data upon collision events.

Preliminary score calculation methods were scripted to process incoming data, laying the groundwork for a future expanded scoring and reward system.

Sprint 7: System Refinement and Stabilization





"System refinement to ensure real-time scoring and robust collision handling"

Overview

With the initial versions of the Food Item attribute system, Trolley detection system, and basic scoring logic in place, sprint 7 was dedicated to refining these core gameplay systems for greater reliability, accuracy, and player satisfaction.

Testing during this phase revealed multiple practical challenges, such as the risk of duplicated collision events when multiple food items entered the trolley simultaneously, inaccuracies in attribute data retrieval, and inconsistencies in score updates.

Efforts were made to streamline data handling, optimize collision detection timing, and improve system feedback, such as updating score displays in near real-time.

This period marked a critical transition from basic functionality to polished usability, ensuring the foundation systems could support later expansions in gameplay complexity.

Objectives

- Refine Food Item attribute assignment and validation
- Improve Trolley collider behavior and collision event handling
- Stabilize scoring logic to prevent miscounts and data loss
- Implement real-time feedback mechanisms for player actions (e.g., score updates)

Implementation Methods

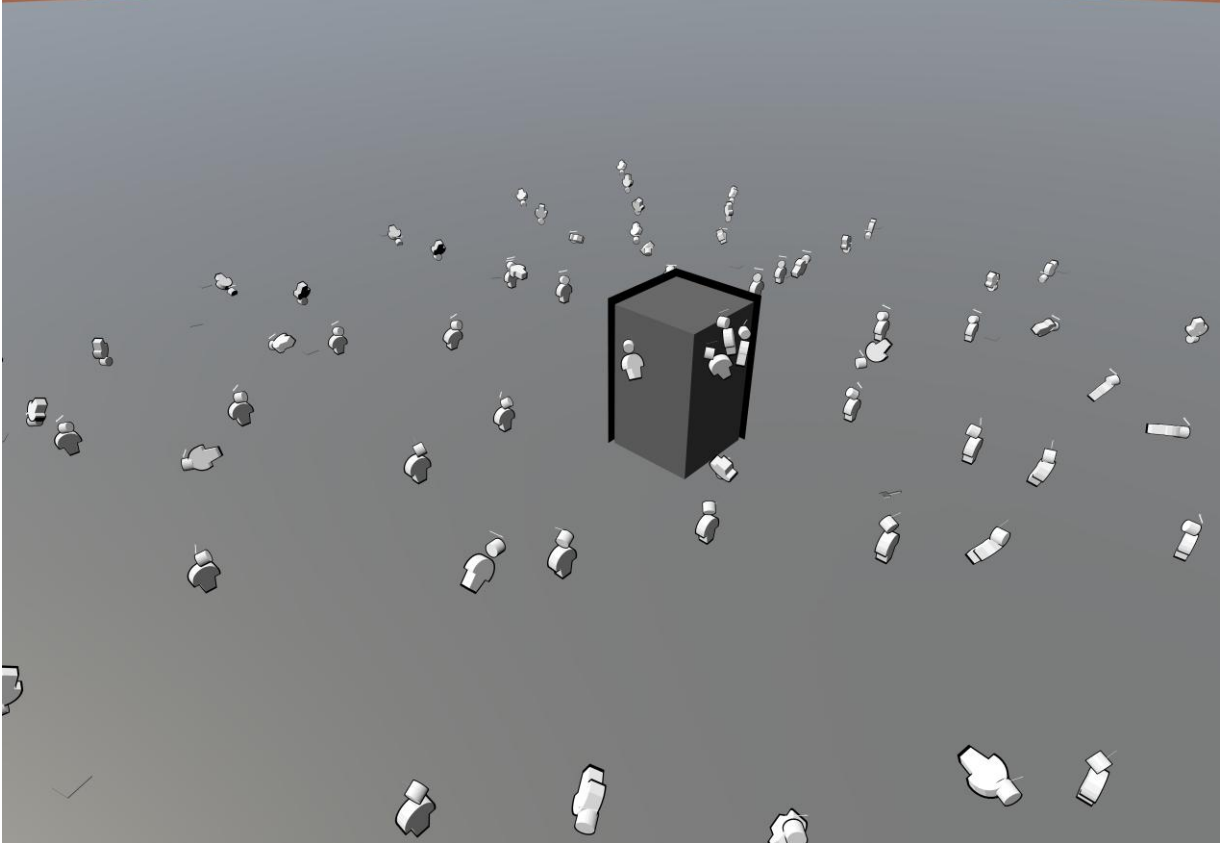
Food Item prefabs were systematically audited to ensure accurate and consistent attribute assignment.

Collision detection scripts were modified to incorporate additional verification layers, preventing multiple triggers from the same item.

Score update functions were optimized to dynamically pull data from detected items immediately upon entry, minimizing frame-delay errors.

Basic UI elements for score display were introduced and linked to the scoring system to provide players with immediate visual feedback during collection.

Sprint 8: Level Expansion and Minimum Viable Product Achieved



“Expansion of further levels and achievement of minimum viable product status”

Overview

Sprint 8 represented a major development milestone, marking the transition from an experimental prototype to a functional Minimum Viable Product (MVP).

The focus during this phase was the creation and integration of a second playable level, expanding the supermarket environment and introducing a more intense pacing of enemy encounters and object interaction opportunities.

Designing the second level required careful balancing of environmental complexity, enemy density, and player task incentives to maintain engagement without overwhelming the player.

This expansion validated the scalability of the previously implemented systems (interaction, scoring, enemy behavior) and demonstrated that the game loop could sustain player interest across multiple stages.

Reaching MVP status meant that the core gameplay vision—survive, interact, and adapt within a semi-chaotic, physically-driven supermarket—was now fully achievable within the VR environment.

Objectives

- Design and construct a new supermarket layout for Level 2
- Adjust enemy spawning mechanics to increase dynamic threat levels
- Integrate Level 2 into the existing scene management and transition logic
- Test and validate full gameplay loop (interaction, threat, scoring) across both levels

Implementation Methods

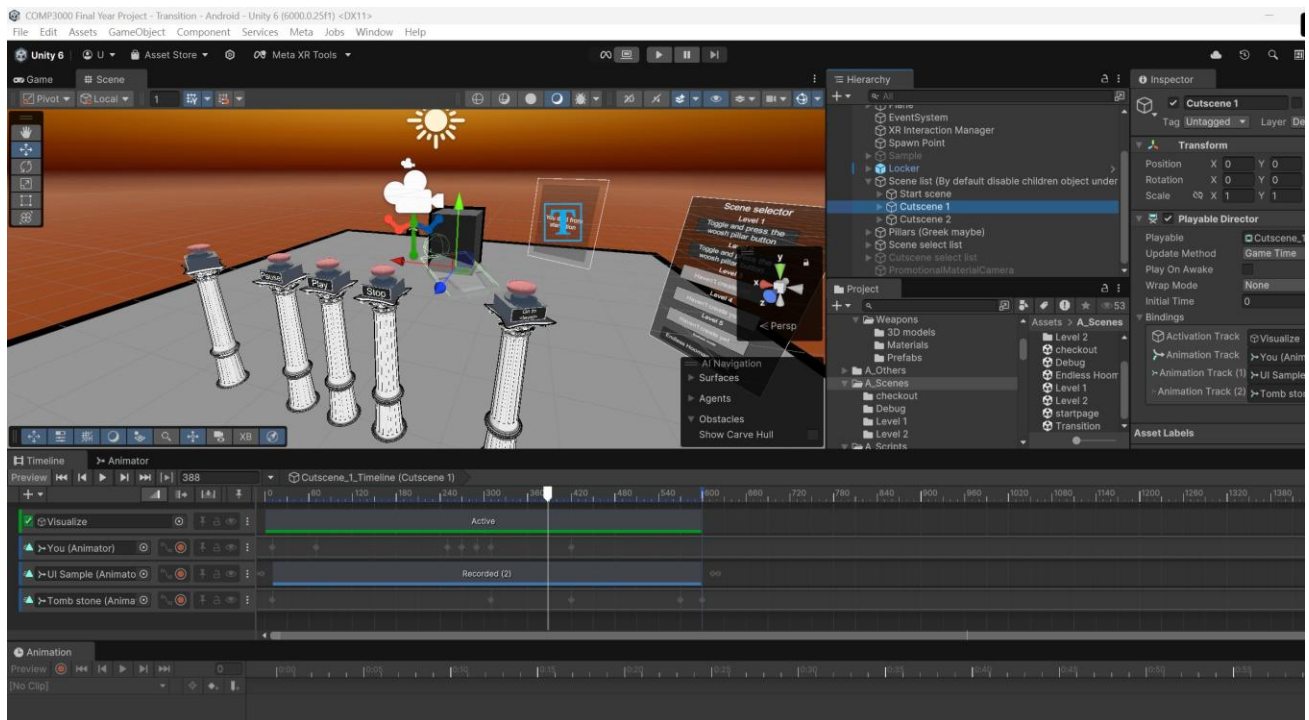
A new section of the supermarket was modelled using modular assets, with altered shelf arrangements and wider aisles to enable varied movement strategies.

Enemy spawn points were redistributed to introduce multi-directional threats, encouraging active spatial awareness.

Scene transition scripts were expanded to manage level loading without disrupting player immersion.

Extensive in-VR playtesting was conducted to fine-tune difficulty scaling, confirm system stability across levels, and verify that the full gameplay loop from collection to confrontation functioned as intended.

Sprint 9: Cutscene System Creation and Narrative Structuring



"Implementation of dynamic cutscene sequences to enhance environmental storytelling"

Overview

Recognizing the need to provide players with a more structured and emotionally resonant conclusion to the gameplay experience, sprint 9 focused on the development of a Cutscape System.

The aim was to create an in-game event manager capable of orchestrating environmental changes—lighting shifts, object animations, enemy spawns—without breaking player immersion or taking away control unnecessarily.

Initial cutscene designs leveraged existing supermarket assets and enemy models, recontextualizing them to create surreal, dreamlike transitions that would foreshadow the ending sequence.

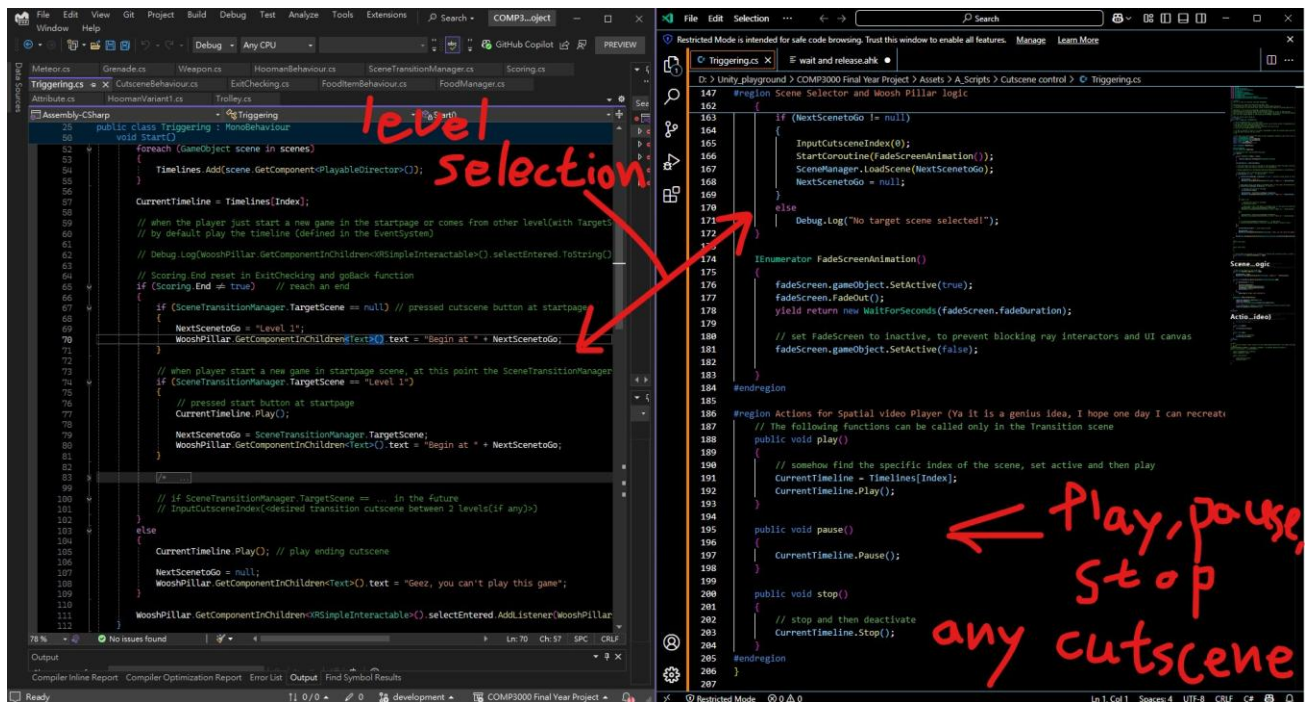
The conceptual inspiration for these cutscenes was drawn from firsthand experiences using the Apple Vision Pro, particularly its emphasis on blending spatial environments and smooth, non-intrusive scene transitions.

This new system laid the groundwork for enhancing narrative delivery within the game without resorting to traditional dialogue or exposition.

Objectives

- Design and implement a modular Cutscape Playback System
- Create initial cutscene sequences using existing assets and environmental effects
- Integrate cutscene triggers into gameplay flow
- Begin conceptualization of the final game ending structure

Implementation Methods



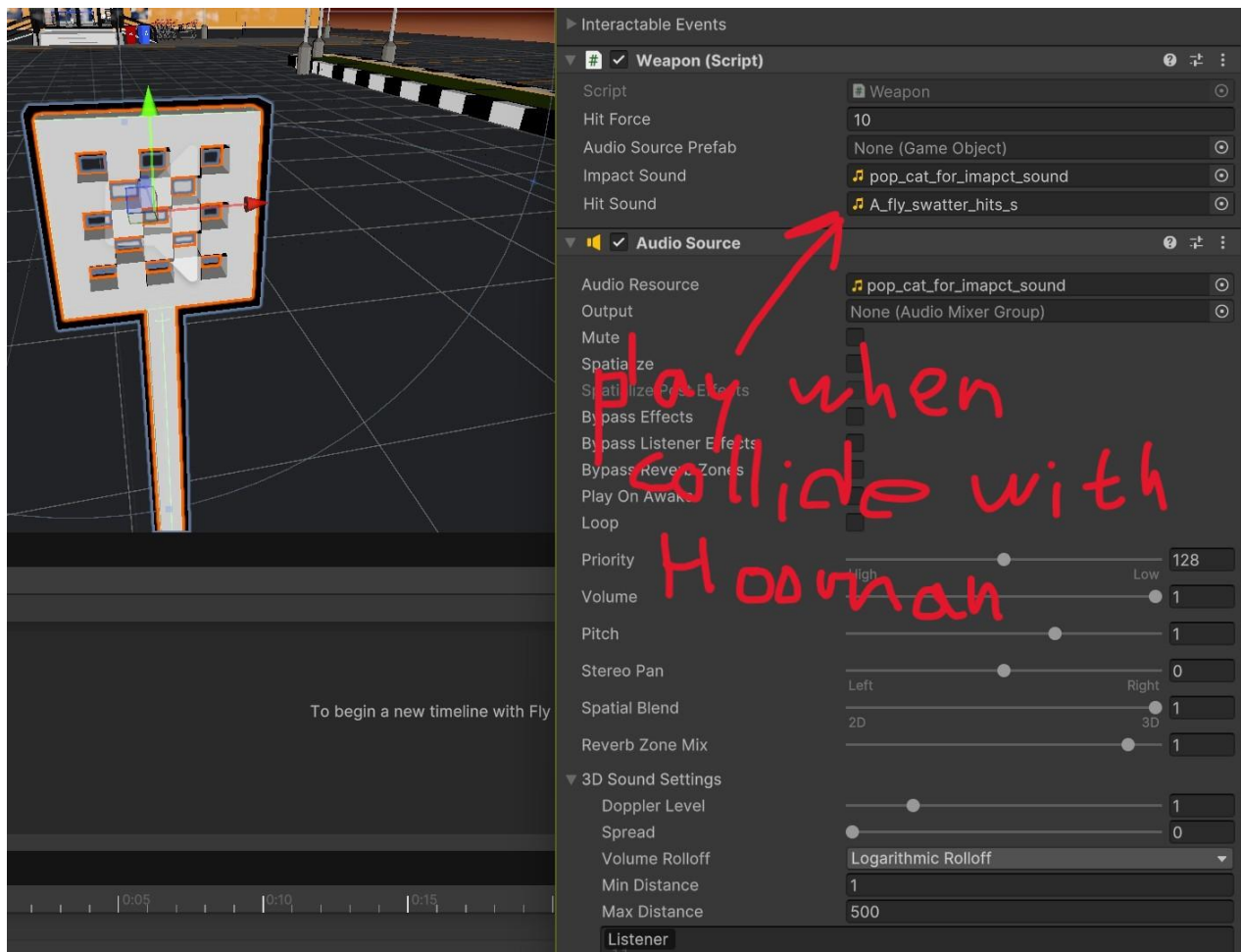
A modular cutscene manager script was developed, capable of sequentially triggering lighting changes, object animations, and enemy events.

Timeline-based events were designed to operate without disabling player input, maintaining VR comfort and immersion.

Existing supermarket assets were rearranged creatively to simulate narrative events such as store collapse, power failures, and enemy surges.

Lighting parameters were tuned dynamically during cutscene playback, drawing directly from spatial transition principles observed in Vision Pro experiences.

Sprint 10: Combat Enhancement, Sound Integration and Playtesting



```
private void OnCollisionEnter(Collision collision)
{
    if (audioSource)
    {
        audioSource.clip = impactSound;
        audioSource.spatialBlend = 1;
        audioSource.Play();
    }

    // Compare if the tag matched Hooman
    if (collision.gameObject.CompareTag("Hooman"))
    {
        // Grab the object's HoomanBehavior script
        hooman = collision.gameObject.GetComponent<HoomanBehavior>();
        if (hooman != null)
        {
            // Hit force calculation
            Vector3 hitDirection = collision.contacts[0].point - transform.position;
            hooman.GetHit(hitDirection, hitForce);

            if (audioSource)
            {
                audioSource.clip = hitSound;
                audioSource.spatialBlend = 1;
                audioSource.Play();
            }
        }
    }
}
```

“Expansion of adding sound effects”



“Enhancement of melee combat and gathering of player feedback through playtesting”

Overview

In sprint 10, major improvements were made to the game's combat feedback and environmental responsiveness.

A new melee weapon—the fly swatter—was introduced, specifically designed to complement the game's physics-driven combat system.

The weapon provided exaggerated but satisfying impact responses, offering players a humorous and highly interactive means of repelling enemies.

To further enhance player immersion and emotional engagement, distinctive sound effects were integrated into the combat system, including impactful "smack" sounds during enemy hits and dynamic collision sounds between objects.

Concurrent with these developments, an informal playtesting session was conducted among peers, gathering initial user feedback.

Players responded positively, highlighting the game's humor, physical engagement, and overall entertainment value, validating the design focus on emergent, playful interaction.

Objectives

- Design and implement a new melee weapon (fly swatter) with distinct physics behavior
- Integrate dynamic sound effects for combat hits and object collisions
- Conduct informal user playtesting sessions for gameplay validation
- Analyze player feedback to inform final polishing phases

Implementation Methods

The fly swatter was modelled and assigned custom Rigidbody and collider settings to ensure lightweight but forceful impacts.

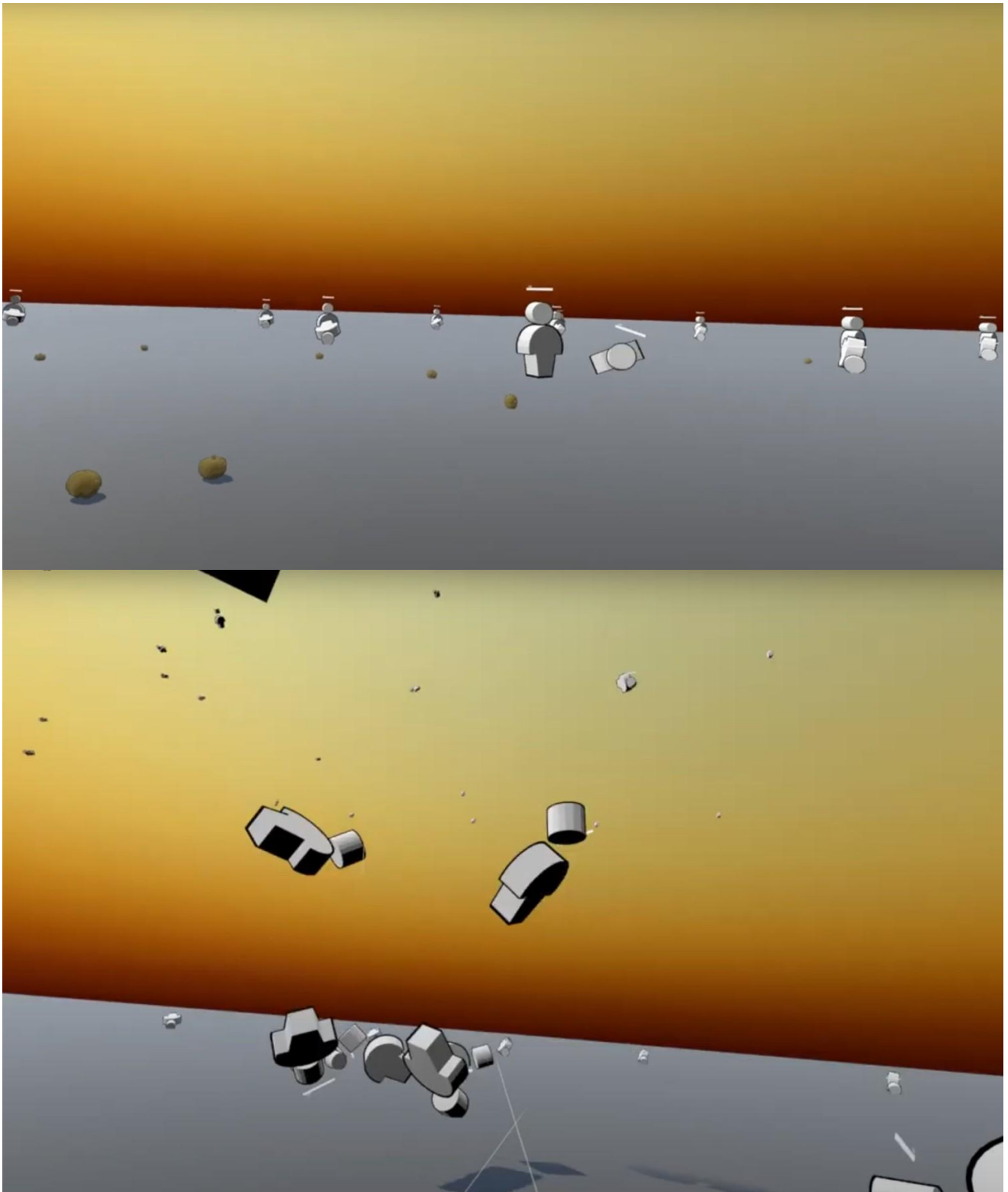
Audio sources were triggered via collision detection scripts, differentiating between melee strikes and environmental object impacts.

Playtesting sessions involved small groups of peers who were encouraged to explore and interact freely; observations and verbal feedback were collected to assess player enjoyment and identify any lingering system inconsistencies.

Sprint 11: Title Screen Redesign and Grenade System Prototyping



“Integration of dynamic title animations”



"Integration of grenade explosion mechanics"

Overview

Sprint 11 marked the final phase of active development, focusing on enhancing the game's initial impression and introducing a new gameplay mechanic.

The title screen was redesigned to feature a dynamic 3D logo that explodes upon game start, creating an immediate sense of immersion and excitement.

This visual effect not only served as an engaging introduction but also laid the groundwork for the grenade system, which was prototyped using similar explosion mechanics.

The grenade system allowed players to interact with the environment in new ways, adding depth to the gameplay.

This period concluded the development cycle, transitioning the project into the final polishing and optimization stages.

Objectives

- Redesign the title screen with a dynamic, exploding 3D logo
- Prototype a grenade system utilizing explosion effects
- Finalize development and prepare for the polishing phase

Implementation Methods

The title screen's 3D logo was constructed using Unity's particle system to simulate an explosion effect upon game start.

This involved configuring particle emitters, adjusting force parameters, and synchronizing sound effects to achieve a cohesive visual and auditory experience.

The grenade system was prototyped by creating a throwable object that, upon collision or after a delay, would trigger an explosion effect similar to the title screen's.

This mechanic introduced new interactive possibilities within the game environment.

With these implementations, the development phase was concluded, setting the stage for final testing and optimization.

Chapter 7: End-Project Report

7.1 Achievements

The project successfully achieved the majority of its original objectives, resulting in a playable and coherent VR experience that demonstrated the intended design philosophy.

Key achievements included:

- **Functional VR Supermarket Environment**
A complete, navigable supermarket space was constructed using modular low-polygon assets, optimized for VR performance.
- **Physics-Based Interaction System**
Players were able to grab, throw, and physically manipulate nearly all objects within the environment using natural hand motions, without reliance on abstract input methods.
- **Dynamic Enemy Behavior**
Enemies demonstrated simple, grounded rolling movement toward the player, with dynamic responses to player actions, including being physically thrown or exploded by force.
- **Explosion and Impact Physics**
A responsive system was implemented where grenades(rotten pumpkin) could create physical shockwaves affecting enemies and props, adding to emergent gameplay possibilities.
- **Environmental Storytelling and Cutscene System**
Trigger-based environmental changes were successfully incorporated, enhancing atmosphere and narrative interpretation without explicit scripting.
- **Optimized Performance for VR Hardware**
Despite heavy use of physics systems, performance was maintained within acceptable framerates on Oculus Quest 2 hardware through careful optimization of asset complexity and physics interactions.
- **Ethical, Legal and Professional Compliance**
The project maintained full compliance with intellectual property guidelines, VR safety best practices, and appropriate depiction of non-graphic content.

7.2 Deliverables Completed

Deliverable	Status
VR Game Prototype	Completed
Physical Interaction System	Completed
Enemy AI Behavior	Completed
Cutscene and Trigger System	Completed
Comprehensive Project Report	Completed

All deliverables outlined at project inception were completed to a satisfactory level.

7.3 Unfulfilled or Partially Fulfilled Goals

Despite achieving the majority of the project's core objectives, several planned features were either partially implemented or left unrealized due to time constraints and shifting development priorities.

These gaps highlight areas for potential future expansion and refinement.

- **Incomplete Environmental Variety**

While the supermarket environment successfully established an interactive and immersive setting, the variety of environmental themes was less extensive than initially envisioned.

Additional zones—such as outdoor parking lots, storage rooms, or back alleys—were conceptualized but ultimately omitted to focus resources on polishing the primary play space.

- **Limited Enemy Behavioral Complexity**

The enemy AI successfully delivered basic pursuit mechanics and reactive physical interactions; however, deeper behavioral variations, such as coordinated group movements or dynamic environmental adaptations, were not implemented.

This limitation slightly reduced the emergent threat complexity, though core engagement remained functional and satisfying.

- **Absence of a Comprehensive Onboarding and Player Guidance System**

One major unfulfilled goal was the development of a structured onboarding system to introduce players to the game's controls, objectives, and environmental interactions.

It was originally intended that players would receive progressive, intuitive exposure to core mechanics—such as object manipulation, scoring rules, and environmental navigation—through embedded tutorials and environmental storytelling techniques.

Due to time constraints and prioritization of system stability, this feature was omitted

from the final build.

As a result, players were required to learn through direct experimentation, which, while potentially enhancing discovery for some, risked alienating less experienced users. The lack of a formal guidance system represents a significant opportunity for future enhancement, especially for improving accessibility and player retention.

7.4 Testing Outcomes

Informal playtesting sessions confirmed the project's success in key experiential goals:

- Players instinctively interacted with objects in natural ways without needing basic movement control instruction.
- Emotional responses such as tension, humor, and discomfort emerged organically through gameplay, participants feels joyful of the gameplay and is interested in further development.
- Physical reactions (enemy rolling, object impacts) were perceived as satisfying and responsive, as well as physical collisions between the trolley and food items, the impact of melee weapon and the enemies.

Some usability issues were observed, including occasional physics instability and minor environment clipping bugs and slight lagging while loading the whole supermarket building with the interior, but none were deemed critically immersion-breaking during final testing phases.

7.5 Overall Assessment

In conclusion, the project achieved its primary goal of creating a physically immersive, emotionally resonant VR experience without traditional narrative structure or violent imagery.

Secondary goals related to AI complexity and environment expansion were deferred but remain clear opportunities for future development.

The final product demonstrates a strong technical and conceptual understanding of VR development, environmental design, and player-centric interaction principles.

Chapter 8: Project Reflections

8.1 Personal Growth and Development

Throughout the course of this project, significant personal and professional growth was achieved.

Moving from initial concept to final prototype necessitated a shift in mindset—from simply "building features" toward **crafting experiences**.

Early assumptions regarding VR interaction underestimated the complexity of designing systems that feel natural and satisfying in a fully immersive environment.

By repeatedly testing, observing player behavior, and iterating on feedback, a deeper appreciation was gained for:

- The subtleties of physics-driven design
- The unpredictability of user actions
- The emotional power of environmental space
- Importance of proper project management flow and schedule of the tasks

The process cultivated a greater sense of patience, critical self-evaluation, and adaptive problem-solving—skills that will prove invaluable in future development work, both within and beyond VR contexts.

8.2 Challenges and Learning Points

Several key challenges were encountered during the project:

8.2.1 Managing Scope in a VR Context

Physics-based VR projects, even seemingly simple ones, are deceptively complex. Every object added, every collider adjusted, potentially introduces stability risks and performance impacts.

Learning to **strictly control scope**—resisting the temptation to endlessly add features—was one of the most important and difficult lessons.

In future projects, scope management will be formalized earlier, with clearer minimum viable product (MVP) definitions and prioritization structures.

8.2.2 Embracing Emergent Design

Initial design documents outlined specific player behaviors and outcomes.

However, live playtesting revealed that players often approached the game in entirely unexpected ways—stacking objects, blocking enemies creatively, or choosing not to fight at all.

Rather than resisting these behaviors, the project adapted to **embrace emergent play**, adjusting systems to support a wider range of valid interactions.

This experience emphasized that **good design empowers player expression** rather than constraining it.

8.2.3 Balancing Realism and Playability

Absolute physical realism, while immersive, can sometimes undermine fun.

For example, early versions of the throwing system were physically accurate but felt sluggish and unsatisfying.

Subtle exaggerations—boosting throw velocity, softening collision detection thresholds—had to be introduced to maintain responsiveness.

Finding this balance between **believability and playability** was an iterative and nuanced process, reinforcing that successful VR design often requires **artful compromise**.

8.3 Strengths Demonstrated

- **Commitment to Iterative Improvement:** Frequent playtesting cycles ensured steady experiential refinement.
- **Technical Flexibility:** Willingness to rebuild or adjust systems when better solutions emerged.
- **Thematic Consistency:** Successfully maintained a cohesive emotional tone throughout the experience without relying on explicit narrative.

8.4 Areas for Future Improvement

- **Formal Playtesting Protocols:** Future projects should incorporate more structured user testing sessions with formal feedback collection instruments (e.g., surveys, interviews).
- **Early Performance Profiling:** Performance optimization should be integrated earlier in the development timeline to avoid late-stage rushes.
- **Greater Accessibility Considerations:** Future VR projects must allocate more resources and design attention toward accessible control schemes and diverse user needs.

8.5 Final Reflection

Ultimately, this project transformed the understanding of what VR can achieve—not merely as a tool for sensory novelty, but as a medium for **emotional engagement, player agency, and silent storytelling**.

The lessons learned extend beyond technical skills, shaping a deeper appreciation for human-centered design principles and the unpredictable beauty of emergent play.

This project stands not only as a completed academic deliverable, but as a foundational step toward a future of more thoughtful, ambitious, and player-empowering interactive experiences.

Conclusions

This project set out to explore the potential of virtual reality as a medium for creating physically grounded, emotionally resonant, and emergent gameplay experiences without relying on traditional narratives or violent spectacle.

Through careful environmental design, intuitive physical interaction systems, and minimalistic enemy behavior, it successfully demonstrated that even mundane settings like a supermarket could be transformed into creative, engaging spaces that provoke joyful player responses.

The development process revealed both the power and the challenges inherent in VR as a creative platform:

- Physics-based interaction can offer unparalleled immersion but demands rigorous balancing between realism and playability.
- Emergent gameplay requires designers to relinquish some control, trusting players to craft their own narratives.
- Environmental storytelling can effectively replace explicit plotlines when space, sound, and interaction are carefully orchestrated.

The project achieved all of its core objectives, delivering a technically functional, thematically coherent, and experientially rich VR prototype.

While certain advanced features—such as adaptive AI behaviors, extended environments, and comprehensive accessibility options—remain aspirational for future iterations, the foundation laid by this work is strong.

Looking forward, the project provides a blueprint for expanding into larger, more complex VR experiences:

- Broader environmental diversity could enrich exploration.
- Multiplayer collaboration could introduce new emergent behaviors.
- Deeper emotional layering through soundscapes and dynamic world states could further heighten immersion.

Ultimately, this project validates the belief that VR's greatest strength lies not merely in simulating reality, but in reshaping it—giving players not just spaces to visit, but **worlds to inhabit, transform, and interpret on their own terms**.

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Appendices

Appendix A – User Guide

A.1 Introduction

This user guide provides detailed instructions for installing, launching, and interacting with the "From Healthy To Zombie" application.

It is intended for players familiar with basic VR device handling, specifically those experienced with the Oculus Quest 2 standalone headset.

The guide outlines system requirements, installation steps, in-game controls, and gameplay objectives to ensure a smooth and enjoyable experience.

A.2 System Requirements

To play "Supermarket Apocalypse VR," the following hardware and software are required:

- **Device:** Oculus Quest 2 VR headset
 - **Software:** SideQuest application installed on a Windows or Mac computer
 - **Storage:** Minimum 500 MB available space on the headset
 - **Internet:** Internet access for downloading SideQuest (not required during gameplay)
-

A.3 Installation Instructions

1. Install the SideQuest application on your PC or Mac (available at sidequestvr.com).
2. Enable Developer Mode on your Oculus Quest 2 via the Oculus mobile app.
3. Connect the headset to your computer using a USB cable and allow USB debugging if prompted.
4. Using SideQuest, locate and install the "Supermarket Apocalypse VR" APK onto the device.
5. After installation, locate the app within the "Unknown Sources" section of your Quest 2 library and launch it directly.

A.4 Controls and Interactions

The core controls within "Supermarket Apocalypse VR" are as follows:

Action	Control
Move	Left joystick
Rotate view left/right	Right joystick (left/right tilt)
Teleport forward	Right joystick (push upward)
Grab objects	Grip button or pinch motion (default Oculus interaction)
Throw objects	Release grip button during hand swing
Attack with held item	Physical swinging motion with the controller

All grabbing and pinching interactions use Oculus Quest 2's default hand gesture settings. No manual configuration is required upon starting the game.

A.5 Gameplay Objectives

The primary objective of the game is to survive within a chaotic supermarket environment overrun by waves of physically-driven enemies.

Players must collect food items, defend themselves using improvised weapons, and strategically manage their movements to maximize their score before reaching the exit.

Key goals include:

- Collect high-value food items based on type and weight.
- Defend against incoming enemy waves using melee or throwable objects.
- Reach the designated exit point with collected items to achieve a final score.

A.6 Troubleshooting

Issue	Solution
Game not appearing in Oculus Library	Ensure the app is installed under "Unknown Sources." Reboot the headset if necessary.
Controller tracking issues	Check for adequate lighting and recalibrate tracking via Oculus settings.
Game performance issues	Close other background apps on the Quest 2. Ensure sufficient device storage is available.
Crashes during gameplay	Restart the application. If persistent, reinstall via SideQuest.

Appendix B – Project management

Sprint 1 – XR Foundation Setup

Objectives:

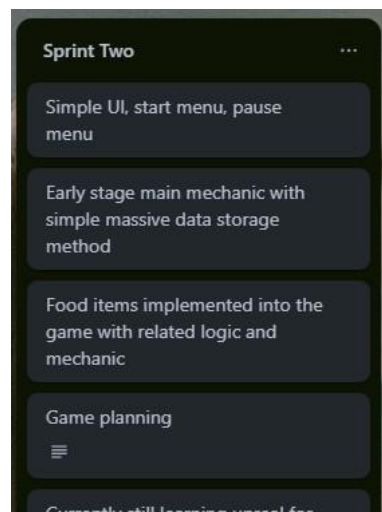
- Set up Oculus Quest 2 XR Rig in Unity
- Configure player height calibration
- Test basic VR movement and head tracking



Sprint 2 – Core Interaction Prototype

Objectives:

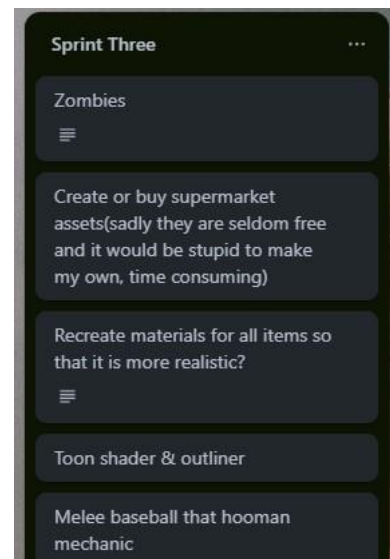
- Implement grab and release mechanics
- Test basic throwable object functionality
- Calibrate hand gesture recognition



Sprint 3 – Enemy Behavior Prototype

Objectives:

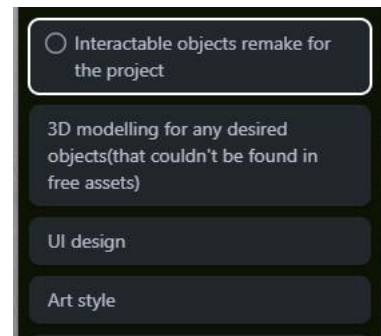
- Develop basic enemy movement toward player
- Implement simple rigidbody-based pursuit
- Enable head rotation to face player



Sprint 4 – Supermarket Environment Construction

Objectives:

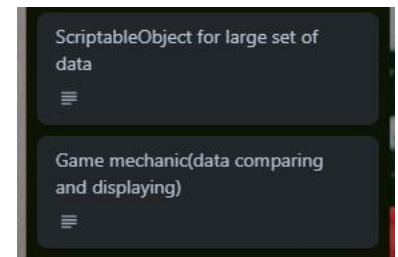
- Design modular supermarket layout
- Populate initial shelves and counters
- Setup colliders for walls and objects.



Sprint 5 – Asset Rework and Food Categorization

Objectives:

- Redesign supermarket assets to match cartoon visual style
- Categorize food items based on type and value
- Populate shelves with categorized food items



Sprint 6 – Attribute System and Trolley Interaction

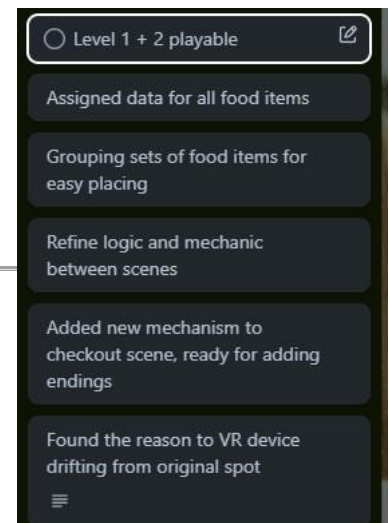
Objectives:

- Implement food attribute classes
- Develop trolley collision and detection system
- Initiate dynamic scoring logic

Sprint 7 – System Refinement and Debugging

Objectives:

- Refine collision detection between food items and trolley
- Prevent duplicate score triggers
- Improve real-time UI score updates



Sprint 8 – Level Expansion and MVP Achievement

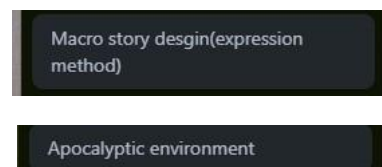
Objectives:

- Create and integrate Level 2
- Adjust enemy spawning and pacing
- Finalize MVP build for full gameplay loop testing

Sprint 9 – Cutscene and Narrative Integration

Objectives:

- Develop modular cutscene trigger system
- Design narrative transition sequences
- Begin planning final game ending



Sprint 10 – Combat and Playtesting Enhancements

Objectives:

- Introduce new fly swatter melee weapon
 - Add impact sound effects
 - Conduct user playtesting and gather feedback
-

Sprint 11 – Final Content and Polishing

Objectives:

- Redesign 3D title screen with explosion effect
- Develop grenade throwing and explosion mechanics
- Prepare project for final submission